Patho Study Guides 1–5 (filled)

NRSG 224 Pathophysiology · Exam 1 covers Classes 1–5 (Tue Oct 13)

Personal study sheets, not for submission.

1 · Cell, part 1
Class study guide, filled from lecture

2 · Cell, part 2 (replication, cancer, genetics)
Class study guide, filled from lecture

3 · Hematopoietic function (Blood)
Luke's own exam-detail notes (made in Grok), converted unchanged

4 · Immune, part 1
Luke's own exam-detail notes (made in Grok), converted unchanged

5 · Immune, part 2 (autoimmunity, inflammation, hypersensitivity)
Class study guide, filled from lecture

Class 1 — Alterations in Cell Function & Growth, Part 1 (The Cell, part 1)

NRSG 224 Pathophysiology · Fall 2026 · filled from the class study guide

> **Personal study sheet — not for submission.** Questions keep the guide's order. Brackets show where each answer came from: S = recorded-lecture slide, mm:ss = recording time (1x), sheet p. = your full study sheet. not covered in lecture = textbook knowledge, not from her lecture.

Sources: recorded lecture (S# = slide, mm:ss = time in the recording), the in-class slides, and your full study sheet pp. 1–4.

Page 1 — The cell: components & functions

Guide prompt: "If pathological changes occur in any of these components → Nursing problems:"

1.1 → problems with nucleus:

- Nucleus = control center → DNA holds instructions for protein synthesis; RNA carries them to cytoplasm [S3–S4, 02:54–06:32; sheet p.1]
- Damaged nucleus → instructions get messed up → can't make certain proteins at all, OR proteins made but defective → don't work [S4, 04:08–06:32]
- Bottom line: trouble with protein synthesis [S4, 04:08–06:32; sheet p.1]

1.2 → problems with ribosomes:

- Ribosomes make proteins (site of protein synthesis out in cytoplasm) [S6, 07:01–07:34; sheet p.1]
- Damaged ribosomes → same problem: proteins not made, or made defective and won't work [S6, 07:01–07:34]
- Class practice Q: "If the function of ribosomes were impaired…" → best answer: "Synthesis of defective proteins, or decreased protein synthesis" [class slide 6]

1.3 → problems with ER:

- ER = system of tubules: (1) transports substances from place to place inside cell, (2) houses ribosomes + enzyme systems (enzymes = catalysts that speed up chemical reactions) [S8, 07:54–09:42; sheet p.1]
- Problems: can't get things place to place; ribosomes have nowhere to hang out → protein synthesis problems; chemical reactions slow way down [S8, 07:54–09:42]
- Class practice Q: impaired ER → best answer: "Substances may not be transported appropriately within the cell" [class slide 7]

1.4 → problems with mitochondria:

- Mitochondria = powerhouse ("Mighty Chondrion") → transforms food energy into ATP; needs oxygen to do it [S11, 10:28–12:06; sheet p.1]
- Not enough O_2 → can't make ATP → power failure → cell can't carry out activities → dies

[S11, 10:28–12:06]

- Her examples: heart attack = heart muscle cells die from power failure; stroke = brain cells [S11, 10:28–12:06]

1.5 → problems with lysosomes (and peroxisomes):

- Lysosomes = bags of digestive enzymes → get rid of worn-out cell parts / foreign material (e.g., bacteria) [S12, 12:06–12:33; sheet p.1]
- Not working → things accumulate inside cell = lysosomal storage disease → cell damaged or malfunctions [S13–S14, 12:33–13:08]
- Example: Tay-Sachs (missing lysosomal enzyme → accumulation in retina + nervous system → blindness, motor/mental issues) [S15, 13:08–14:15]
- Peroxisomes = specialized lysosomes that detoxify/get rid of free radicals (unstable atom/molecule that can damage cells) → free radical damage blamed for wrinkles, cancer, atherosclerosis, heart disease [S19, 15:16–16:36; sheet p.1]
- ★ **Prof emphasis:** she won't test on Tay-Sachs specifics — "more likely to ask" what happens if lysosomes aren't working [S18, 14:43–15:16]

1.6 → problems with Golgi complex:

- Golgi = post office of the cell → takes ribosome-made proteins, packages them into secretory vesicles/granules, directs them to the right place [S20–S21, 16:36–17:55; sheet p.1]

- Not working → proteins made don't get to the right place → can't do their job [S21, 17:06–17:55]
- Example: I-cell disease — lysosome enzyme packaged/released into blood → lysosome missing enzyme (storage disease) + enzyme breaks down things in blood you still want → damage to heart, lungs, nervous system, MSK deformities [S22–S23, 17:55–20:36]
- ★ **Prof emphasis:** no test questions specifically on I-cell disease — only the underlying pathology of lysosome/Golgi malfunction [S23, 19:23–20:36]

1.7 → problems with cell membrane:

- Membrane holds cell together — keeps what belongs inside in, keeps out what doesn't; carrier/transport proteins + ion channels let O_2/fuel in and waste out [S24, 20:36–21:35; sheet p.1]
- Receptor sites (hormones/chemical messengers) → cellular communication; damaged receptors → poor communication AND medications that stimulate/block receptors won't be effective [S26, 22:11–24:26]
- Self-antigens tell immune system "self vs non-self" → if damaged, immune system may attack self-cells = autoimmune disease [S26, 22:11–24:26]
- Also maintains resting membrane potential + generates action potentials; worst case: cell breaks apart and dies [S26, 22:11–24:26]

Page 2 — Cell metabolism & movement across the membrane

2.1 Nursing problem (altered cellular metabolism/energy production):

- Cellular metabolism = cell makes energy: food → ATP, stored until needed [S27, 24:26–25:26]
- Two pathways: aerobic (requires O_2, done in mitochondria) vs anaerobic (no O_2, less efficient, fewer ATP, but a good backup) [S27–S29, 24:26–26:26; sheet p.2]
- Anaerobic too long → lactic acid → changes body fluid pH = acidosis → lots of things don't function normally [S30, 26:26–27:53]
- 3 things cell needs for energy: fuel, oxygen, appropriate enzymes (catalysts) → missing any one → not enough energy → cells malfunction or die [S30, 26:26–27:53; sheet p.2]

2.2 Pathological change in movement across membrane → Nursing problem:

- 3 ways across membrane: passive movement, active transport, endocytosis/exocytosis [S31, 27:53–29:00; sheet p.2]
- Passive = down concentration gradient (high → low), no energy, goal = equilibrium; diffusion = particles move, osmosis = water moves; deciding factor = membrane permeability [S32–S43, 29:00–34:32]
- Hypernatremia example: too much Na^+ in blood → bigger gradient → Na^+ still can't cross → MORE water osmoses out → cell shrinks →

brain cells shrink = lots of neuro issues [S44–S47, 34:32–37:04]

- Active transport = against gradient, requires energy (ATP) — Na^+/K^+ pump keeps K^+ in, Na^+ out → needed for action potentials [S48–S49, 37:04–40:38]
- Endocytosis (pinocytosis = fluid/dissolved "cell drinking"; phagocytosis = solids, e.g., neutrophils/monocyte-macrophages eating bacteria) → problem = big impact on immunity; exocytosis releases hormones/gets rid of waste → problem = can't release hormones [S50, 40:38–42:58]

Page 3 — Cellular communication & membrane potentials

3.1 Nursing problem (alterations in availability/function of chemical messengers or receptor sites):

- Chemical messenger method: one cell releases hormone/neurotransmitter/growth factor → binds receptor site on 2nd cell → tells that cell what to do [S51, 42:58–46:13; sheet p.2]
- Two places it breaks: (1) too much or too little messenger, (2) damaged or missing receptor sites → message never gets delivered [S51, 42:58–46:13]
- Her examples: Parkinson's = not enough dopamine (tremor, slow stiff movements); schizophrenia = too much dopamine; myasthenia gravis = autoimmune attack on receptors at neuromuscular junction → muscle weakness [S51–S52, 42:58–47:34]

- Predisposing factors per sheet: autoimmune dz, congenital defects, genetic predisposition, degeneration d/t age or toxin exposure [sheet p.2]
- Meds work by increasing/decreasing messenger or stimulating/blocking receptors [S52, 46:13–47:34; sheet p.2]

3.2 Nursing problem (changes in membrane permeability):

- Resting membrane potential: cell electrically polarized, inside more negative (K^+ diffuses out, Na^+ can't get in → mostly anions left inside) [S54–S58, 49:37–52:59; sheet p.3]
- Action potential: something changes membrane permeability → Na^+ diffuses in → inside becomes positive = depolarization; ions then pass cell-to-cell through gap junctions (nerve/muscle communication) [S59–S61, 52:59–55:25]
- Repolarize (2 steps): membrane goes back to not letting Na^+ in + Na^+/K^+ pump pumps K^+ back in, Na^+ back out → ready to repeat [S62–S64, 55:25–57:00]
- Problems: electrolyte imbalance messes up the gradient; hypoxia/impaired metabolism (no O_2 or glucose) → no ATP → no Na^+/K^+ pump → can't repolarize → brain & heart can't function [S66, 57:19–59:30; sheet p.3]

Page 4 — Cell injury: adaptive responses, reversible injury, cell death

Guide lists each response with space under it. Fill: what it means clinically, upside/downside, and her example(s).

4.1 atrophy

- Cells get smaller / revert to lower level of functioning when not enough O_2, fuel, or blood flow → smaller cell needs less O_2/nutrients → more likely to survive [S79, 74:33–76:03; sheet p.4]
- Upside: cell survives, doesn't die. Downside: not as functional [S79, 74:33–76:03]
- Her examples: calf muscle atrophy from chronic impaired arterial blood flow → can't do as much work; atrophied endocrine gland → less hormone [S80, 76:03–76:47]

4.2 hypertrophy

- Cells get bigger in response to increased work demands — opposite of atrophy [S81, 76:47–77:05; sheet p.4]
- Examples: bodybuilders lifting heavier weights (skeletal muscle cells); left ventricle in chronic high BP (pushing against high arterial pressure → heart muscle cells hypertrophy) [S82–S83, 77:05–77:56]
- Upside: cells can do more work. Downside: bigger cells doing more work need more blood, O_2, fuel → if atherosclerosis limits supply → may set person up for a heart attack [S84, 77:56–78:40]

4.3 hyperplasia

- Increase in cell NUMBER (not size) — usually in response to chemical stimulus: change in hormone or growth factor
  [S85, 78:40–80:31; sheet p.4]
- Examples: wound healing (growth factor → more cells to fill in wound); pregnancy uterus = BOTH hypertrophy + hyperplasia
  [S85, 78:40–80:31]
- Downside: tissue enlarges enough to interfere with surrounding structures — benign prostatic hyperplasia squeezes urethra shut → trouble emptying bladder [S86, 80:31–81:29]
- ★ **Prof emphasis:** "students do mix those two terms up a lot on the test" — hypertrophy = size, hyperplasia = number [S85, 78:40–80:31]

4.4 metaplasia

- Cells change from one cell type to another that's better able to survive — triggered by chronic irritation/inflammation
  [S87, 81:29–82:19; sheet p.4]
- Not a change out of primary type (not nerve→muscle); e.g., one kind of epithelial cell → a sturdier one [S87, 81:29–82:19]
- Her example: smoker's airway ciliated columnar epithelium → stratified squamous (sturdier, less likely to be injured)
  [S88, 82:19–83:14]
- Downside: lose protective cilia → mucus + trapped bacteria/irritants get stuck in airway, can't get out [S88, 82:19–83:14]

4.5 dysplasia

- Book's definition she likes: "deranged cell growth" — cells can be all different shapes and sizes, don't really look like each other [S90, 83:45–84:27; sheet p.4]
- Usually a reaction to chronic irritation/inflammation; progression: normal → metaplasia → dysplasia if irritation isn't removed [S90–S91, 83:45–86:00]
- No upside; big downside = precursor to cancer (higher risk, NOT a guarantee) [S91, 84:27–86:00]
- Examples: airways of smokers; uterine cervix — Pap smear finding dysplastic cells ≠ cancer, means higher risk → more frequent Paps [S91, 84:27–86:00]

4.6 reversible injury

- Cellular response ladder depends on severity of stressor: adapt → become injured → die [S78, 74:10–74:33; sheet p.3]
- If the stress is severe the cell becomes injured anyway; if we get rid of the source of the stress, cell may heal and injury be reversed [S92, 86:00–89:15]
- If injury is bad enough, the cell will actually die (→ 4.7) [S92, 86:00–89:15]

4.7 cell death/necrosis – if cell injury is irreversible

- How bad depends on how many cells die: a few liver cells = probably fine; a whole bunch

= liver failure → patient may die
[S92, 86:00–89:15]

- Replacement varies: skin cells replace themselves all the time; brain cells do so at a much, much slower rate [S92, 86:00–89:15]
- Dead cells replaced either by same cell type OR by scar tissue — heart attack: heart muscle cells replaced with scar tissue → restores structure but NOT function (no action potentials, no contraction) → chronic heart failure [S92, 86:00–89:15]
- Lab markers of cell damage: draw blood, look for proteins/enzymes normally found only inside cells → high levels = cells damaged, insides leaking out. Troponin → heart muscle cells (chest pain in ED) [S74, 69:15–70:51]
- CK and LD → "damage to the liver or the skeletal muscles or some tissues like that" — she does not pair them one-to-one (your sheet writes CK → skeletal muscle, LD → liver)
[S74, 69:15–70:51; sheet p.3]

Class 2 — Alterations in Cell Function & Growth, Part 2 (Cell replication, cancer & genetics)

NRSG 224 Pathophysiology · Fall 2026 · filled from the class study guide

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Sources: recorded lecture audio, transcribed by machine (S# = slide, mm:ss = time in the recording), the in-class slides, and your full study sheet pp. 5–7.

Page 1 — Cell replication/differentiation → Cancer

Guide: Mutations in regulatory genes → abnormal proliferation + abnormal differentiation. Example of disorder: cancer. Each printed finding has space under it; fill with what it means / what you'd assess.

1.1 → uncontrolled cell growth

- More cells actively cycling; fewer in the resting phase → uncontrolled growth.
  [S10, 08:59–12:15]
- Lose cell density-dependent inhibition; do not die within their normal programmed time frame. [S20, 28:54–30:09]
- Increasing tumor cells crowd out normal cells and steal their blood supply.
  [S21, 30:09–30:33; sheet p.6]

1.2 → infiltration into surrounding tissues

- Infiltration = growth extending into surrounding normal tissue; malignant tumors lack a containing capsule. [S38, 50:34–51:10]
- Loss of cell-to-cell cohesiveness → cells break away and travel to other body sites: metastasis. [S37, 47:31–50:34]
- Infiltration makes complete surgical removal difficult; remaining cancer cells keep multiplying → recurrence.
  [S37, 47:31–50:34; S38, 50:34–51:10]

1.3 → tissue/organ dysfunction

- Abnormal differentiation → cancer cells do not specialize properly or perform normal tissue functions. [S22, 30:33–31:33]
- Normal cells are crowded out while abnormal cells do not function normally → tissue/organ dysfunction. [S21, 30:09–30:33; S22, 30:33–31:33]

1.4 → changes in cell characteristics

- Cells fail to differentiate/specialize properly → different characteristics and impaired normal function. [S22, 30:33–31:33; S23, 31:33–32:12]
- Lose normal cell-to-cell cohesiveness → break away and metastasize. [S37, 47:31–50:34]
- May produce abnormal substances and display abnormal cell-membrane antigens.
  [S24, 32:12–32:31; S29, 36:43–37:09]

1.5 abnormal biochemical products (ex: VEGF, ectopic hormone production, toxins, etc.)

- VEGF = vascular endothelial cell growth factor → new blood vessels supply oxygen/nutrients, allowing continued tumor growth.
  [sheet p.6; S25, 32:31–33:37]
- Ectopic hormones: cancer cells make hormones without normal negative-feedback regulation → excess hormone/endocrine-like disease. [S26, 33:37–35:49]
- Enzymes/toxins damage tissue → sores that do not heal normally. [S27, 35:49–36:32]
- Abnormal antigens such as PSA and CEA can help monitor cancer progression and treatment response; professor describes this role rather than diagnosis/screening.
  [S29, 36:43–37:09; S30, 37:09–40:49]

1.6 abnormal antigens

- Abnormal differentiation → abnormal antigens on cancer-cell membranes; these can serve as clinical markers. [S29, 36:43–37:09]
- Immune system normally fights cancer; immune defects/immunosuppression increase cancer risk. [S18, 25:49–27:10; S44, 58:23–61:14]
- Examples of increased risk: AIDS, immunosuppressive drugs, and aging with less efficient immune function. [S18, 25:49–27:10]

1.7 generalized effects

- Cachexia = weight loss; WBC cytokines cause anorexia/loss of appetite → reduced eating

and weight loss. [S33, 42:10–42:53; S34, 42:53–43:48]

- Pain: tumor pressure on nerves, inadequate blood/oxygen, and tissue damage from enzymes/toxins. [sheet p.6; S35, 43:48–45:34]
- Anemia: poor nutrition (iron/B-vitamin deficiency), bleeding from damaged tissue, or tumor erosion into vessels. [S36, 45:34–47:31]
- Bone-marrow involvement impairs RBC production; cancer treatments can also cause anemia. [S36, 45:34–47:31]

1.8 Example of disorder: benign neoplasms — Nursing problems:

- Benign = noncancerous: abnormal proliferation only, with normal differentiation; malignant = both abnormal. [S37, 47:31–50:34]
- Usually fibrous capsule → contained expansion, no infiltration or metastasis; easier to remove surgically. [S39, 51:10–51:43; sheet p.6]
- Nursing problems are mainly local effects of increasing size/location—not the abnormal substances made by cancer cells. [S40, 51:43–53:39]
- Can still be fatal if growth interferes with a vital function; benign brain tumors are an example. [S40, 51:43–53:39]

1.9 Interventions (cancer / neoplasms) — blank Interventions column

- Surgery: remove tumor; infiltrating cancer is harder to remove completely, and remaining cells can regrow. [S37, 47:31–50:34; S41, 53:39–54:47]
- Radiation: generates free radicals that damage cancer cells/interfere with replication;

rapidly proliferating cells are most affected, but normal cells can be injured.
[S42, 54:47–55:56; class slide 20]

- Chemotherapy interferes with the cell cycle: S-phase drugs inhibit DNA replication; M-phase drugs prevent division. [S10, 08:59–12:15]
- Different phase targets explain combination drugs and treatment long enough for cells to reach the targeted phase. [S10, 08:59–12:15]
- Chemo effects are not confined to cancer → side effects; cancer treatment may cause anemia. [S43, 55:56–58:23; S36, 45:34–47:31]
- Biologic response modifiers/immunomodulators modify immune responses: monoclonal antibodies and synthetic cytokines. [S44, 58:23–61:14]

1.10 Pathological change: Mutations in regulatory genes — which genes? (proto-oncogenes/oncogenes, tumor-suppressor genes) and PF details

- Genetic mutations alter programming that controls proliferation and differentiation; specific proto-oncogenes/oncogenes and tumor-suppressor genes are not named in these sources. [S13, 15:56–18:11; S14, 18:11–20:45]
- Carcinogens: chemicals, radiation (including sun), cigarette exposure; free-radical DNA damage can cause mutations. Exposure-to-diagnosis lag may be 5–30 years.
[S15, 20:45–23:10; sheet p.6]
- Cancer-causing viruses listed: HPV, EBV, HBV; incorporation of viral DNA can alter host-cell

instructions → abnormal replication.
[sheet p.6; S17, 24:08–25:49]

- Immune defects: AIDS, immunosuppressive drugs, aging → increased cancer incidence. [S18, 25:49–27:10]
- Heredity: germ-cell mutations can pass to children; mutations in regular body cells do not. Cancer risk usually reflects a combination of factors. [S19, 27:10–28:54]

Page 2 — Genetic aspects of the cell

2.1 Gene mutation → Nursing problem:

- DNA replication error/mutation → altered instructions for a protein; radiation or toxic exposure may contribute. [S10, 08:59–12:15; S61, 78:08–79:26]
- Protein is absent or defective/nonfunctional → nursing problems depend on that protein's role. [S62, 79:26–82:02]
- Severity ranges from mild to fatal; no single set of findings applies to all single-gene disorders. [S62, 79:26–82:02]
- Germ-cell mutation can be passed to the next generation. [S61, 78:08–79:26]

2.2 Sex-linked: hemophilia A (factor VIII) — nursing problems/findings

- Hemophilia A: deficiency/nonfunction of factor VIII → assess muscle hematomas and hemarthrosis (joint bleeding). [sheet p.7]
- X-linked recessive pattern: males with a defective X usually have disease; females with

one defective X are usually carriers.
[S66, 84:28–85:47; S67, 85:47–87:03]

- ★ **Prof emphasis:** sex-linked test questions specify a son or daughter—calculate risk using the two possibilities of that sex, not all four children. [S84, 98:48–100:21]

2.3 Autosomal dominant: Marfan's syndrome — nursing problems/findings

- Autosomal dominant: one defective gene causes disease; Aa = disease, aa = normal; no carriers in the taught pattern.
[S72, 88:29–89:57]
- Defective connective tissue → long, thin arms and fingers; musculoskeletal deformities.
[S73, 89:57–90:47; S75, 91:09–91:49]
- Assess eye problems: retinal detachment or lens dislocation from weak connective tissue/ligaments. [S75, 91:09–91:49]
- Assess cardiovascular problems: valve prolapse and aneurysms. [S75, 91:09–91:49]

2.4 Autosomal recessive: sickle cell anemia — nursing problems/findings

- Autosomal recessive: two defective genes needed; aa = disease, Aa = carrier, AA = normal genes. [S76, 91:49–93:44]
- Abnormal RBCs clump together → obstruction/clot-related problems.
[S77, 93:44–95:07]
- Hemolytic component: breakdown of RBCs/hemoglobin contributes to anemia.
[S78, 95:07–95:58]

2.5 Chromosomal disorders: Trisomy 21 (Down syndrome) — findings; also mitosis vs meiosis errors (nondisjunction)

- Mitosis: somatic/body cells; meiosis: germ cells, with 23 chromosomes per egg/sperm → 46 at fertilization. [S4, 01:32–02:10; S9, 07:55–08:59]
- DNA-copying errors can cause gene disorders; chromosome-separation errors cause unequal chromosome numbers. [S10, 08:59–12:15]
- Meiotic nondisjunction: one daughter cell receives both copies, the other none → after fertilization, trisomy or monosomy. [S95, 117:31–118:28; S96, 118:28–119:10; S98, 120:06–120:38]
- Down syndrome = trisomy 21: three copies of chromosome 21. [S99, 120:38–122:13]
- Findings: characteristic facial features, intellectual disability, possible congenital heart problems and GI malformations. [S99, 120:38–122:13]
- Structural errors can also occur: a chromosome piece breaks off/is lost → loss of multiple genes. [S93, 114:36–116:33]

2.6 Interventions (genetic disorders) — blank Interventions column

- Single-gene interventions depend on the affected protein and problems caused; no universal treatment. [S62, 79:26–82:02]
- Chromosomal disorders: treat symptoms, maximize quality of life, and help families support development. [S100, 122:13–124:03]

- Genetic counseling: Punnett squares estimate the chance of an affected child, carrier, or child with normal genes. [S80, 96:17–97:13]
- ★ **Prof emphasis:** Punnett squares will be on the test—diagram parents, fill boxes, label each child's status, then calculate the specific risk asked. [S81, 97:13–97:41]
- For her sex-specific X-linked questions, use X/Y and risk out of 2; for autosomal questions, use A/a and risk out of 4.

  [S84, 98:48–100:21; S88, 106:29–108:06; S90, 110:58–112:26]

Patho Study Guide 3 — Blood (Hematopoietic function)
Luke's own notes (made in Grok), converted from his Pages file with the wording unchanged. Personal study sheet, not for submission.

PATHOPHYSIOLOGY — NURSING

Hematopoietic Function

NRSG 224 — Blood / Recorded Lecture 3

Complete exam-detail notes from the PowerPoint lecture recording. Slide content plus what the instructor said out loud — definitions, numbers, mechanisms, clinical applications, and every comparison she flagged.

1. Hematopoietic function — the big picture

Hematopoietic function = how the body makes blood cells. This lecture walks the three cell lines that come from bone marrow stem cells, then what goes wrong when you have too many, too few, or poorly working cells in each line.

She said (not on the slide): She opened by asking: what three kinds of cells does bone marrow make? Answer them by BOTH names.

Line	Formal name	Everyday name	Main job
Red cells	Erythrocytes	RBCs	Carry O2 (via hemoglobin / iron)
White cells	Leukocytes	WBCs	Defend against infection / foreign material
Platelets	Thrombocytes	PLTs	Primary hemostasis — platelet plug

TEST TIP: If a question asks for the three products of bone marrow, give both names. She used erythrocytes / leukocytes / thrombocytes on purpose.

2. Hematopoiesis and differentiation
All three lines come from hematopoietic stem cells in red bone marrow. New cells pass through levels of differentiation until they become mature circulating cells.

Process name: hematopoiesis.
Regulated by growth factors — she named erythropoietin (EPO) and granulocyte colony-stimulating factor (G-CSF).

She said (not on the slide): *The reason she mentioned growth factors is clinical: we can now make synthetic versions in the lab and you will give them. Synthetic EPO-type drugs for anemias; G-CSF when neutrophils are too low.*

She then said: focus on red cells first. The rest of the lecture follows RBC → hemostasis/platelets/factors → WBC → leukemia (WBC cancer).

3. Red cells, hemoglobin, and erythropoiesis

Job of the RBC

Transport oxygen to tissues. That works because of iron inside hemoglobin. The slide shows the hemoglobin molecule with iron binding O2.

Erythropoiesis feedback loop — memorize the chain

She asked: what is the PRIMARY stimulus for erythropoiesis?

Hypoxia (not enough O2 at the tissues) → kidney makes EPO → bone marrow makes RBCs → more O2 carrying capacity.

She said (not on the slide): The diagram also showed the scale of production: on the order of 200 × 10⁹ RBCs per day. Lungs pick up O2, tissues take it.

TEST TIP: Hypoxia is the stimulus. The kidney is the sensor/EPO factory. The marrow is the factory floor. Do not skip a link.

Reticulocytes — almost-mature RBCs

RBCs leave the marrow slightly immature = reticulocytes. The developmental path on the slide: hemocytoblast → proerythroblast → … → reticulocyte → erythrocyte.

She said (not on the slide): At any one time about 1% of circulating RBCs are reticulocytes. That is why we use a retic count as an index of marrow activity.

Retic count	What it means	Possible reasons she gave
Lower than normal	Marrow is not making new RBCs as fast as usual	Missing building blocks (nutrition) OR marrow problem / suppression. Count does NOT tell you WHICH cause.
Higher than normal	Marrow is pumping out RBCs faster than usual (more still slightly immature)	Most common trigger she named: blood loss. Marrow tries to replace what was lost.

TEST TIP: **Low retic = lazy or blocked marrow (or no ingredients). High retic = marrow is trying. After hemorrhage you EXPECT a high retic if the marrow works.**

5. Blood Cells — When they get old

When blood cells get old or defective they get phagocytized, where some of amino acids and iron is saved to be reused. Part of hemoglobin that doesn't get recycled gets converted to bilirubin. Bilirubin get transported to liver by albumin. The liver than excretes the bilirubin in the bile. The bile goes into the GI tract and out in the stool. Bilirubin is what gives poop its brownish yellowish color

4. Blood types — ABO and Rh

ABO

RBCs have surface antigens used to group people. She walked the table out loud.

Type	Antigen on RBC	Antibody in plasma
A	A	Anti-B
B	B	Anti-A
AB	A and B	Neither
O	Neither	Anti-A and Anti-B

She said (not on the slide): ***You are NOT born with these ABO antibodies. Babies spontaneously develop them by about 3–6 months of age WITHOUT being transfused the***

wrong blood — probably because similar antigens exist in the environment.

Wrong-type transfusion: antigen meets its antibody → hemolysis (RBC bursting). That is why the bursting-cell slide sits right after ABO. When destroyed cells go through kidney they get stuck in small glomurulli and Cause damage.

Rh

Named for the rhesus monkey (antigens discovered 1940). Many antigens (C, D, E); D is the most reactive and is used for Rh typing. Rh+ has the D/Rh antigen; Rh− does not.

DO NOT MIX UP: ABO antibodies appear spontaneously. Rh-negative people make anti-D ONLY if they are actually exposed to Rh-positive blood. Never exposed → never make the antibody.

If an Rh− person is exposed (wrong transfusion or an Rh− mother carrying an Rh+ fetus), they can then make anti-D. Later exposures can destroy Rh+ cells.

5. Polycythemia — too many RBCs
Polycythemia = too many red cells. Two predisposing factors on the slides, built one at a time.

1. Bone marrow disorder

Example: polycythemia vera — a type of blood cancer. Marrow itself overproduces RBCs (and

often other lines).

2. Chronic hypoxia — she called this the MUCH more common reason

***She said (not on the slide):* Hypoxia tells the kidney to make EPO (erythropoietin). If you are always a little hypoxic, the kidney is always making extra EPO, so the marrow always makes extra RBCs.**

Causes of chronic hypoxia she listed:

- Smoking
- Chronic lung disease such as COPD (chronic obstructive pulmonary disease)
- Living at high altitude — less O2 in the air

She said (not on the slide): *Elite athletes exploit this: train at altitude → more RBCs → hope to carry more O2 to muscle → more endurance.*

Effects — two bullets, both testable

Effect	Why it matters
↑ Blood viscosity	Blood is thicker. Harder to push. Higher clot (since blood likes to clot when still) / sluggish-flow risk.
↑ Blood volume	More cells = more volume. Stresses the heart and vessels. BP^

6. Anemia — not enough working RBCs / hemoglobin
Slide contrast: normal smear vs anemic smear (fewer cells). Anemia = lower than normal RBCs or hemoglobin, so tissues get less O2.

How patients feel / look (symptom slide)

Fatigue, weakness, dizzy, pale skin, rapid heartbeat, shortness of breath, chest pain, headache, leg pain, insomnia. These are the O2-delivery failure picture — not a separate disease.

Four predisposing factors — she built the list across many slides

Memorize all four. Questions often give a scenario and want the CATEGORY.

#	Category	High-yield examples from the recording
1	Nutritional deficiencies	Iron; B12 (cobalamin); B9 (folic acid)
2	Bone marrow failure	Aplastic f — empty / fatty marrow, not making cells
3	Blood loss	GI bleed, trauma, postop hemorrhage
4	Excessive / premature destruction (hemolysis)	Sickle cell (from last unit); toxins; burns; mechanical valves; autoimmune attack on RBCs

7. Nutritional anemias and the CBC

Iron deficiency → microcytic, hypochromic anemia

Slide pair: normal peripheral smear vs microcytic hypochromic anemia. Cells are small (microcytic) and pale (hypochromic) because they lack hemoglobin/iron.

How you know without looking in a microscope — CBC (complete blood count test)

She said (not on the slide): You look at MCV and MCHC on the CBC, not just the hemoglobin.

Index	Stands for	What it tells you	In microcytic hypochromic iron deficiency
MCV	Mean corpuscular volume	SIZE of the RBC	DOWN (smaller cell → less volume)
MCHC	Mean corpuscular hemoglobin concentration	How much Hb is packed in each cell	DOWN (less Hb → lower concentration)

TEST TIP: Low MCV + low MCHC = microcytic hypochromic picture. She walked it as a Socratic question: if the cell is smaller, does volume go up or down? Down.

She said (not on the slide): CBC also reports hemoglobin, RBC count, platelets, WBCs — but MCV/MCHC are the ones that classify the SIZE and COLOR of the anemia.

B12 (cobalamin) and B9 (folic acid) → macrocytic / megaloblastic

These vitamins are needed to build DNA in rapidly dividing marrow cells. Without them, RBCs come out LARGE (macrocytic) and often megaloblastic (big, immature-looking). Retic path slide is reused to remind you where maturation can stall.

Vitamin	Also called	Classic extra problem besides anemia
B12	Cobalamin	Neuro changes (myelin)

B9	Folic acid / folate	Neural-tube defect / spina bifida in the fetus

B12 absorption and pernicious anemia

B12 needs intrinsic factor (IF) to get from gut into blood. IF is made by parietal cells in gastric mucosa.

She said (not on the slide): B12 is a delicate vitamin. Digestive enzymes in the GI tract can destroy it. IF protects B12 until it can be absorbed.

Ways she listed to lose IF:

- Chronic gastric inflammation / gastritis — stomach not making enough IF
- Part of the stomach removed — cancer surgery or weight-loss surgery — parietal cells gone
- Certain drugs interfering with production or activity of IF

Pernicious anemia = anemia because there is not enough IF, so B12 cannot be absorbed. She was very specific: ONLY use that name when IF is the reason B12 is missing.

TEST TIP: B12 deficiency from vegan diet or from ileal disease is still B12-deficiency anemia. It is pernicious anemia only when IF is the problem.

B12 deficiency — neuro changes (come from lack of intake, lack of absorption, or you are missing something in order to absorb vitamin B12 (intrinsic Factor (IF))

She said (not on the slide): B12 is also needed to make normal myelin — the sheath around axons that speeds conduction. No normal myelin → neuro issues.

Term	What she said it means
Paresthesias	Numbness, tingling, odd sensations
Ataxia	Coordination problems
Position sense / vibratory sense changes	Sensory deficits — you cannot tell where a limb is in space as well

TEST TIP: Folate deficiency does NOT give this neuro package. That contrast is high-yield even if she did not put it as a bullet.

B9 deficiency — spina bifida (come from lack of intake or lack of absorption)

Especially worried about pregnant women. Vertebrae do not close in the back → opening. Spinal nerves or even cord can protrude.

She said (not on the slide): Degrees of severity on the picture: (1) bony defect only, nothing sticking out; (2) meninges ± spinal nerves sticking out; (3) actual cord sticking out. If nerves/cord are outside, they can be damaged → paralysis.

Practical point she hit: make sure pregnant patients get their supplements.

Cheilosis / cheilitis (caused by either iron or Vitamin B deficiencies)

Cracked, inflamed corners of the mouth. Shown as a clinical clue that can travel with B-vitamin deficiency anemias.

8. Bone marrow failure (reason for amemia)

Aplastic anemia slide: marrow not producing the cell lines. Red cells (and often WBC + platelets) drop. Causes later overlap with thrombocytopenia and neutropenia: radiation, toxic drugs/chemicals, marrow cancer.

9. Blood-loss anemia

Acute bleed picture on the GI-bleed slide: pumping, oozing, exposed vessel, red or black clot.

Acute blood loss cells look normocytic normochromic — normal size, normal color — because you lost whole blood. You did not lose the ability to build hemoglobin inside each cell. MCV and MCHC stay normal at first.

TEST TIP: Chronic slow GI bleed can later look iron-deficient (microcytic hypochromic) once iron stores are spent. Acute hemorrhage starts normocytic.

Hypovolemia → shock — longest spoken segment in the lecture

Especially with a LARGE loss in a SHORT time, the problem is not only anemia. It is low circulating volume.

She said (not on the slide): Word breakdown: hypo = low, vol = volume, -emia = blood → low blood volume.

Why postop / post-procedure vital signs are frequent (she gave a schedule): maybe every

15 minutes the first hour, then every 30 minutes the next hour — to catch bleeding.

Physiology she taught	What you OBSERVE first
Lost volume → BP wants to fall	Heart rate rises FIRST (compensation: pump remaining blood into arteries faster)
Tachycardia can keep BP normal for a while	BP still normal + climbing HR = already worry
When HR can no longer compensate	THEN you see the BP drop
BP so low organs are not perfused	Shock → multi-organ failure → can die

***She said (not on the slide):* Simplistic definition for this course (detail later in critical care): shock = BP has dropped to the point that you can no longer adequately perfuse major organs and they start to fail.**

TEST TIP: Patho logic: volume down → BP down. Bedside observation: tachycardia before hypotension. If you wait for the BP to fall, you are late.

10. Hemolytic anemia

Fourth cause: excessive or premature RBC destruction.

***She said (not on the slide):* Callback to last class: sickle cell — sickled cells look defective so the body destroys them = hemolytic anemia.**

Other destroyers she listed:

- Toxic chemicals or bacterial toxins
- Physical trauma — severe burn damages RBCs in that area
- Artificial heart valves — even flexible valves can smash RBCs when they close

Autoimmune disease — immune system attacks your own RBCs

Breakdown of hemoglobin → bilirubin → jaundice (yellow). That is why jaundice slides sit next to hemolysis and bile-path diagrams. Extra bilirubin dumped into the bile/liver pathway.

Symptoms of anemia: signs of hypoxia(tired, weak, short of breath, pale). Inc heart rate,

11. Hemostasis — how bleeding stops

She said (not on the slide): She said her study guide lists FIVE stages of clot formation. She went through them in order.

Stage	What happens	Triggered by / purpose
1. Vessel spasm / vasoconstriction	Injured vessel clamps down	Injury + prostaglandins (e.g. thromboxane A2) from injured cells. Limits blood leaving the hole.
2. Platelet plug (primary hemostasis) (stored in spleen for 8 hours b4 release)	Platelets stick to wall and to each other	Contact with injured wall + prostaglandins. Plugs the hole but is UNSTABLE — flow can knock it loose.
3. Fibrin clot (secondary hemostasis)	Coagulation cascade lays fibrin over the plug	Stabilizes the plug so it stays put.
4. Clot stays while vessel heals	Fibrin mesh holds	Cascade end-product is fibrin, not the platelet itself.
4 1/2.. Clot Retraction	Liquid gets squeezed out of fibrin clot, pulling edges of wound together	
5. Clot degradation / fibrinolysis	Plasmin breaks fibrin down	Checks and balances — do not leave clots forever.

DO NOT MIX UP: **Platelets ≠ clotting factors. Platelets = marrow, make the plug. Clotting factors = LIVER, make the fibrin clot that stabilizes the plug. She repeated this later and told you not to mix them up.**

Intrinsic vs extrinsic cascade (part of fibrin clot)

Pathway	Activated by	Spoken example
Intrinsic	Injury to the BLOOD VESSEL	Atherosclerosis can injure the vessel without injuring surrounding tissue — so the two paths are not totally redundant
Extrinsic	Injury to TISSUES	Cut, trauma — tissue factor

She said (not on the slide): ***She used to think two pathways were redundant (how do you injure a vessel without injuring tissue?). Atherosclerosis is the exception. Newer research: more overlap than textbooks first taught — still teach both until they know more.***

Both pathways start on different factors, then JOIN. End result of either path: fibrin clot. Factors circulate in INACTIVE form and sequentially activate the next one (cascade).

She said (not on the slide): ***Checks and balances: blood also has anticoagulant / anti-clotting factors so you only clot where you need to, not everywhere.***

12. Platelet problems — thrombocytopenia

Example she used: thrombocytopenia = too few platelets. Bleeding because you cannot build a stable plug.

Predisposing factors (built across slides)

Factor	Why
Bone marrow depression	Marrow makes platelets. Radiation, toxic drugs/chemicals, marrow cancer → fewer platelets out.
Splenomegaly	Spleen = large. After release, platelets hang in the spleen up to ~8 hours. Big spleen sequesters more platelets AND over-clears old/defective cells, including platelets.
Autoimmune destruction	Antibodies attack platelets (ITP-type picture).

What bleeding looks like

Epistaxis = nosebleed (slide title).
Petechiae = pinpoint bleeds in skin (photo slides).
Easy bruising in general when plugs cannot form.

Interventions she named when the cause is autoimmune

Splenectomy — take out the organ that is sequestering / eating platelets.
Immunosuppressants — quiet the immune attack.
Plasmapheresis / plasma exchange — machine looks like dialysis. Pull blood, separate plasma from cells, give cells back with synthetic plasma, throw away the patient's plasma because that is where the attacking antibodies live. Two or three rounds can put some patients into remission. Not every autoimmune disease responds.

Platelets that exist but do not WORK — antiplatelet drugs

***She said (not on the slide):* Before leaving platelets she added a whole spoken section that is barely a title on the next slide: people can bleed with a NORMAL platelet COUNT if the platelets cannot clump.**

Aspirin is the prototype antiplatelet. It stops platelets from clumping into a plug → bleed longer, bruise more.

Same property is used ON PURPOSE: prevent arterial clots that cause ischemic stroke / TIA / stable angina / ACS / PAD / MI. (All terms for heart issues or lack of blood flow)

***She said (not on the slide):* Mini-dose / baby aspirin once a day, or one adult aspirin every other day, is NOT for headaches or cramps. One baby aspirin will not treat pain. If you see that dose, it is for antiplatelet effect.**

Slide table of antiplatelet classes (know they exist and that bleeding is the shared adverse effect):

Aspirin — GI bleed, hemorrhagic stroke risk.
P2Y12 blockers: clopidogrel (Plavix), prasugrel (Effient), ticlopidine (Ticlid) — ACS / stent / stroke / PAD; Ticlid also thrombocytopenia, neutropenia, TTP.
PDE inhibitors: dipyridamole (Persantine) after valve replacement with warfarin; cilostazol (Pletal) for claudication.
GP IIb/IIIa inhibitors (ReoPro, Integrilin, Aggrastat) — potent; GI / urinary / retroperitoneal bleeding.

13. Clotting-factor problems

Two directions on the slide: too many factors / too much clotting, or not enough.

Too many / too much clotting → thromboembolic events

Thrombus = clot where it formed. Embolus = clot that traveled. Together: thromboembolic events (MI, ischemic stroke, PE, DVT picture).

Not enough factors — four predisposing factors

Factor	Mechanism she taught
Liver disease	Liver MAKES clotting factors.(don't get confused with platelets that are made by bone marrow) Sick liver → not enough factors. Also may not make bile.
Vitamin K deficiency	Many factors need vit K to be made in a WORKING form. Factors may be produced but they do not function.
Hereditary disorders	Classic: hemophilia (Factor VIII was the example in the genetics unit).
Increased consumption	Used up faster than replaced — lead-in to DIC. (Disseminated Intravascular Coagulation)

How you become vitamin K deficient (spoken, not all on the slide):

Diet low in vit-K foods.
Cannot ABSORB it: vit K is fat-soluble → needs BILE. Liver disease → no bile → no vit K absorption → now you fail to make working factors TWO ways (liver cannot synthesize them AND cannot absorb K).
Broad-spectrum antibiotics (BSA) — the slide after vit K. Gut bacteria help make vit K. Wipe them out with BSA → less K → worse clotting.

TEST TIP: Platelets = marrow. Factors = liver + vitamin K. BSA hits the bacterial K source, not the marrow.

DIC — disseminated intravascular coagulation

***She said (not on the slide):* Break the words: coagulation = clotting; intravascular = inside vessels; disseminated = widespread. So: clotting in vessels all over the body.**

NOT a primary disease. It is a complication of another critical illness: severe infection, multiple trauma, advanced cancer — ICU patients. Inflammation / chemical mediators of inflammation are thought to flip the cascades on.

Two problems at once:

Microthrombi occlude flow → tissue ischemia → organ failure.
Making all those clots CONSUMES the clotting factors → nothing left when you need to stop a bleed → hemorrhage.

She said (not on the slide): *Paradox: abnormal clotting AND abnormal bleeding at the same time. Some patients show only one side.*

TEST TIP: DIC is the example of “increased consumption” on the factor-deficiency list.

14. White blood cells

She said (not on the slide): She got confused in nursing school because one cell can have several names. She tried to collapse the

aliases. There are not as many types as it feels.

Three MAIN groups: granulocytes, monocytes, lymphocytes.

Granulocytes — granules visible under the microscope

Cell	Also called	Job from the recording
Neutrophil	Polymorphonuclear leukocyte / PMN; mature = segs; immature = bands	Phagocyte. First-line eater of bacteria / foreign material. Lysosomal enzymes digest what it engulfs.
Basophil	A granulocyte	She grouped it with the other two granulocytes on the lineage slide (allergy / histamine role is the usual pairing — lecture emphasis was the trio).
Eosinophil	A granulocyte	Primary job: parasites (too big to phagocytose). Granule enzymes damage parasites. Also pile up in allergy / asthma. Some enzymes pro-inflammatory, some anti-inflammatory — likely different subtypes. Body's checks and balances.

Segs vs bands and "shift to the left"

CBC often splits neutrophils into segs and bands.

Segs = MATURE neutrophils. Nucleus looks segmented.

Bands = IMMATURE neutrophils. Nucleus is a horseshoe / band.

She said (not on the slide): *Why "left"? Old lab slips listed bands on the LEFT and segs on the RIGHT. More bands = shift toward the left column.*

Modern slips stack them top/bottom but the phrase stuck.

Shift to the left = more bands than usual. Most common cause she named: infection (or inflammation). Marrow speeds neutrophil production so more slightly immature cells spill out. Also called neutrophilia (too many neutrophils).

TEST TIP: Left shift on a CBC is a clue the marrow is responding to infection, not a random lab quirk.

Monocytes and lymphocytes (preview of the immune lecture)

Monocytes: phagocytes in BLOOD. When they leave the vessel into tissue they are called macrophages or dendritic cells. More next class.
Lymphocytes — adaptive immunity, more next class:
B lymphocytes mature in Bone marrow → become antibody producers → humoral immunity.
T lymphocytes mature in Thymus → cell-mediated immunity.
NK (natural killer) cells — innate immunity.

Too few WBCs — the pinea words

She said (not on the slide): Suffix -penia = deficiency. Prefix a- = absence / nearly without.

Term	Means	How worried
Leukopenia	Too few leukocytes (WBCs in general)	General term
Neutropenia	Too few neutrophils	High concern — neutrophils are the most plentiful WBC and first phagocytes on scene

Granulocytopenia	Too few granulocytes (often used like neutropenia)	Same neighborhood
Agranulocytosis	Neutrophils so low it is almost like having none	She used a number: neutrophil count < 200. Normal frame she gave: about 3,000–7,000. Below 200 ≈ without granulocytes

Neutropenia — predisposing factors (same logic as platelets)

Factor	Spoken explanation
Bone marrow problems	Marrow makes neutrophils. Same insults: radiation, drugs, chemicals, marrow cancer.
Increased destruction (autoimmune)	Immune system attacks neutrophils.
Splenomegaly	Spleen clears old/defective blood cells. Big spleen over-clears neutrophils too.
Increased consumption	Used up faster than replaced. Neutrophils migrate into a bad wound/infection and leave the bloodstream. A raging infection can drain circulating neutrophils faster than marrow can refill.

Assessment / interventions slide was a prompt for you to think: infection risk, fever, mucous membranes, protective precautions — and drugs that stimulate production.

G-CSF (granulocyte colony-stimulating factor) — the synthetic growth factor she flagged at the start of the lecture. Give it to push the marrow to make neutrophils.

15. Review: cancer applied to blood — leukemia

She rebuilt the cancer list from the previous unit, one bullet per slide, then applied it to WBC cancers.

Cancer rule from last unit	How it shows up in leukemia
Cells don't function properly	Blasts / poorly differentiated WBCs cannot do mature immune jobs
Cells crowd out normal cells	Marrow packed with blasts → anemia + thrombocytopenia + neutropenia together
Can metastasize	Leukemic cells already live in blood; infiltrate organs
Causes anorexia / cachexia	Same two mechanisms as last unit: steal nutrients + cytokine anorexia
Causes pain	Marrow expansion, infiltration, ischemia, tissue damage

Blasts. (Lymphocytic leukemia specifically cancerous changes in lymphocytes or their precursors, myelogenous leukemia interferes more quickly with more types of blood cells. Its when the stem cell is cancerous.

Very immature white-cell precursors. Acute leukemias are blast-heavy. Poorly differentiated — callback to the differentiation ladder from Cell Part 2.

Acute vs chronic

Acute: sudden, many blasts, poorly differentiated, sick fast.

Chronic: more mature-appearing cells, slower course (still cancer).

Healthy vs leukemia marrow diagram: healthy side has stem cells → fully differentiated immune cells. Leukemia side is crowded with poorly differentiated cells.

Therefore you could have acute/chronic lymphocytic leukemia and acute/chronic myogenic leukemia.

How we look at the marrow

***She said (not on the slide):* Aspiration = withdraw fluid. Biopsy = bigger needle, actual chunks of marrow. Usual site: iliac crest. Alternatives: tibia; sternum if those cannot be used. Sternum looks scary — periosteum is tough; they push hard. She watched one at the bedside.**

Goal of the specimen: see how differentiated the cells are, stage the leukemia, name the specific type.

Stem-cell / bone-marrow transplant (process slide)

Collection from patient or donor blood/marrow → lab processing to purify/concentrate → cryopreservation (freeze) → patient gets high-dose chemo and/or radiation to wipe marrow → thawed cells reinfused. Allogeneic = donor cells.

16. Rapid-fire test bank

Marrow makes erythrocytes, leukocytes, thrombocytes.
Hematopoiesis is growth-factor driven (EPO, G-CSF); we give synthetic versions.
RBC job = O2 via iron in hemoglobin.

Erythropoiesis chain: hypoxia → kidney EPO → marrow RBCs.
Retic ~1% of RBCs. Low retic = sluggish marrow or no ingredients. High retic = marrow responding; most common trigger = blood loss.
ABO antibodies by 3–6 months without transfusion. Rh antibodies only after exposure.
Polycythemia: PV (marrow cancer) vs secondary from chronic hypoxia (smoke, COPD, altitude, athletes). Effects = viscosity + volume.
Anemia 4 causes: nutrition, marrow failure, blood loss, hemolysis.
Iron: microcytic hypochromic; MCV↓ MCHC↓.
B12/B9: macrocytic/megaloblastic. Pernicious = no IF (parietal cells / gastritis / gastrectomy / some drugs). B12 neuro = myelin (paresthesia, ataxia, position/vibration). B9 = spina bifida.
Acute bleed: normocytic normochromic. Watch HR first, then BP. Shock = organs not perfused.
Hemolysis: sickle, toxins, burns, mechanical valves, autoimmune → jaundice.
Hemostasis: spasm → platelet plug (unstable) → fibrin clot (cascade) → plasmin later. Intrinsic = vessel; extrinsic = tissue; they join.
Platelets ≠ factors. Thrombocytopenia: marrow, big spleen (8-hour hangout + over-clearing), autoimmune. Treat autoimmunity with splenectomy, immunosuppressants, plasmapheresis.
Aspirin / antiplatelets: bleed more; used to prevent arterial clots. Baby aspirin = antiplatelet dose, not analgesia.
Factor deficiency: liver, vit K (diet, no bile, BSA), hereditary, consumption/DIC.
DIC: ICU complication; microclots + used-up factors = clot AND bleed.
Neutrophil = PMN = segs (mature) / bands (immature). Left shift = infection. Eosinophils = parasites + allergy. Mono in blood = macrophage/dendritic in tissue. B/T/NK preview.

Leukopenia vs neutropenia vs granulocytopenia vs agranulocytosis (<200 neutrophils vs normal ~3,000–7,000).

Neutropenia causes: marrow, autoimmune destruction, spleen, consumption into infected tissue. Drug: G-CSF.

Leukemia = blood-line cancer using the five cancer rules. Blasts. Aspirate vs biopsy. Transplant steps.

17. End-of-deck cartoons (not test content)

Slides 94–105 are cartoons (leukocyte / platelet / iron / pernicious-definition jokes) and a “doctor wants more blood tests” closer. Use them as a mental break, not as sources of new mechanisms.

Pernicious on those last slides is the dictionary joke (destructive / deadly) — a memory hook for why they named IF-related B12 anemia “pernicious.”

Patho Study Guide 4 — Immune system, part 1
Luke's own notes (made in Grok), converted from his Pages file with the wording unchanged. Personal study sheet, not for submission.

PATHOPHYSIOLOGY — NURSING

Immune System — Part 1

NRSG 224 — Recorded lecture 4. Exam-detail notes from the recording. Slide bullets plus what she said that never appeared on the slide. Red text is added from her posted study-guide handout.

She said (not on the slide): *This is a pre-class lecture. The immune system is complicated enough to fill a quarter; she is compressing it into two class periods and wants a few things remembered rather than a long list forgotten.*

1. Four functions of the immune system

She collapsed the job list to four. The first is the one people already know. The other three are the ones students skip.

Function	What she wants you to say
Immunity	Resistance to disease by defending against pathogenic organisms.
Self vs non-self	Recognize foreign molecules and get rid of them.
Inflammation	Not just a bad feeling. It is a tool to defend against harmful agents and start healing.
Cancer surveillance	Callback to the cancer lecture: abnormal cells probably arise all the time; the immune system is supposed to recognize and clear them.

She said (not on the slide): ***These are not four separate machines. The same self-vs-non-self mechanism is used to spot a cancer cell and to spot a pathogen. Lots of overlap.***

Self vs non-self works because almost every cell has surface antigens with unique molecular shapes. White cells have learned which shapes belong to you. Her handout name for those shapes is antigenic determinant sites, or epitopes. She contrasts autoantigens (normal body markers) with foreign antigens.

Antigen vs antibody — she stopped the lecture for this

DO NOT MIX UP: Antigens and antibodies are the two words she sees mixed up most often on the test.

> Antigen = a marker on a cell membrane used to decide self vs non-self. Antigens are not automatically bad. A self antigen is left alone. A foreign antigen is targeted.
> Antibody = what the immune system MAKES to get rid of an antigen it has decided is foreign.

TEST TIP: Antigen in and of itself is neither good nor bad. The interpretation (self vs foreign) decides.

2. Two components of the immune system

Immune cells

Fancy name for white blood cells, plus the substances they make: antibodies and cytokines. She pointed back to the blood lecture for the cell types.

Lymph structures — primary vs secondary

Kind	Organs she named	Job	If it fails
Primary	Bone marrow and thymus	Where immune cells are produced or mature	Impairs immunity a lot, especially bone marrow
Secondary	Spleen, lymph nodes, tonsils, appendix, patches in GI, respiratory, and reproductive tracts Her handout names the GI patches Peyer's patches, and the mucosal tissue MALT (mucosa-associated lymphoid tissue) in respiratory, GI, and reproductive systems. Role she wrote: trap antigens and promote their interaction with mature immune cells.	Clean foreign material out of lymph before it returns to blood	Less efficient, but people lose tonsils or appendix all the time and do not get sick constantly

TEST TIP: Primary = born/matured there. Secondary = filters. Losing a secondary organ is not the same as losing bone marrow.

3. Two major immune mechanisms

	Innate	Adaptive
Also called	Natural / native / nonspecific	Specific / acquired
When ready	Present at birth, jumps in right away	Acquired later; not ready at birth
How picky	Anything foreign. Not specific.	Learns the particular pathogens you meet

Speed vs power	First line. Holds pathogens until adaptive kicks in, and helps activate adaptive.	Slower to start, more effective because it is aimed at one antigen

***She said (not on the slide):* Together they are a complete package: innate covers you immediately against anything foreign; adaptive takes longer but does not spread its attention thin.**

4. Five components of innate immunity

1. Skin and mucous membranes

***She said (not on the slide):* Intact barriers keep pathogens out. Preventing infection is more efficient than clearing one that already started. A break in the skin loses this first line.**

2. Phagocytic white cells

Mainly neutrophils and monocyte-macrophages. Two jobs:

Engulf and digest foreign particles, tumor cells, and virus-infected cells, holding the line until adaptive immunity starts.
Act as antigen-presenting cells (APCs). They process the foreign antigen and display it on their membrane so a T cell can see it. That display is what activates adaptive immunity.

3. Natural killer (NK) cells

***She said (not on the slide):* NK cells are the only lymphocyte in innate immunity. T cells and B cells are adaptive.**

Job: kill tumor cells and virus-infected cells. Not by phagocytosis. They release toxins or enzymes that damage the target.

4. Cytokines

Expanded definition: a hormone-like protein released by certain white cells that acts on other white cells to regulate the immune response. A white cell telling other white cells what to do.

Examples she named: interferons, interleukins, tumor necrosis factor, colony-stimulating factor.

What they can do (her study-guide list is examples, not complete):

- Increase or decrease production, maturation, and activity of white cells (CSF / growth factors).
- Mediate inflammation.
- Produce fever. Handout mechanism: cytokines act on the hypothalamic thermoregulatory center and raise the set point.

She said (not on the slide): *Fever is useful. Many pathogens cannot survive the higher temperature, and fever makes some white cells more active. Some cytokines do more than one job; functions overlap. Clinical reason she cares: we now give synthetic cytokines, and we have drugs that*

block cytokines. If you know the job, you can figure out why the drug is ordered.

5. Complement system

***She said (not on the slide):* She compared it to the clotting cascade: both are proteins sitting in blood in inactive form; each factor activates the next. End of clotting = clot. End of complement = a more efficient immune system, plus inflammation. Her handout: a cascade of about 20 plasma proteins.**

Job: complement the work of phagocytes and antibodies in clearing pathogens.

TEST TIP: If she lets you remember only one word with complement, she said that word is inflammation.

What activation does	Spoken meaning
Mast cells and basophils degranulate	Mast cells are very similar to basophils. Granules hold mediators of inflammation: histamine, leukotrienes, bradykinin. Degranulation = inflammation.
Enhanced chemotaxis	Chemical signal that calls white cells to the area that needs them.
Lysis	Bursts pathogens or abnormal cells, including cells that have become cancerous.
Opsonization (a process)	Coats antigen-antibody complexes with complement pieces (green balls on her slide) so phagocytes can grab them. Many bacteria have a capsule that is hard to phagocytize. Antibody gives a handle; complement makes the handle even easier to pull in.

5. Adaptive immunity — T cells and B cells
Provided by T and B lymphocytes. Each T or B cell is programmed for ONE specific foreign antigen. That is the difference from innate.

Branch	Cell	Targets	How
Cell-mediated	T cells	Microbes INSIDE cells — usually viruses	Cannot enter the cell either, so it kills the infected cell
Humoral	B cells	Microbes OUTSIDE cells — usually bacteria	Antibodies. Antibodies cannot go inside cells, so this branch is for extracellular targets

***She said (not on the slide):* Bacteria can sometimes go inside cells. If they do, T cells take that job too.**

Helper T (CD4) — the master switch

CD4 is the membrane marker used to identify this T cell. Almost all white cells have markers; CD4 is the helper's marker.

Called helper because, once an APC(antigen presenting cell) shows it the antigen it was programmed for, it releases cytokines that activate the other T cells and the B cells.

***She said (not on the slide):* The diagram also shows it influencing macrophages, neutrophils, and dendritic cells. It does NOT need to activate innate immunity. It DOES need to activate the rest of adaptive immunity. It still makes innate cells work more efficiently. Wipe out helper T and**

adaptive immunity does not turn on, and innate immunity works worse. That is why she calls it the master switch.

Cytotoxic / killer T (CD8)

Kills tumor cells and virus-infected cells by releasing toxins and enzymes — same method as NK cells. Two differences she made you name:

Cytotoxic T is adaptive, so it must be activated by helper T first. NK is innate and does not need helper T.

Cytotoxic T targets one specific foreign antigen. NK attacks anything foreign.

Suppressor / regulatory T

Two jobs:

Slow or stop the immune response once the pathogen is gone. Checks and balances so the response does not run on.

Suppress self-reactive white cells (cells that would attack normal self). That is how it helps prevent autoimmune disease. Her handout adds that regulatory/suppressor T cells are believed to be a subset of CD4+ or CD8+ cells.

Memory T

Hangs around a long time and remembers that pathogen. You do not need a large standing army of helper and cytotoxic T cells for an infection that is already gone, but a

second exposure should be recognized fast enough to mobilize before you get sick.

***She said (not on the slide):* She read that memory T cells may live longer because they have more mitochondria, so they can make more ATP. Cute cartoon: the memory cell telling E. coli "I'll never forget you" means next time I will kill you.**

Humoral side — plasma cells and memory B

B cells also must be activated by helper T first, then specialize.

> Plasma cells make antibodies. Antibodies bind surface antigens and act like handles so phagocytes can grab encapsulated bacteria. Antigen + antibody = antigen-antibody complex. Coat that with complement (opsonization) and clearance is even easier.
>
> Memory B hangs around. On re-exposure it quickly drives proliferation of plasma cells so a large antibody burst happens before you get sick the second time.

6. Five antibody classes

Class	Slide line	What she added
IgG	Most plentiful; crosses placenta	Only one that crosses the placenta. Gives newborns most of their protection for the first few months.
IgA	Secretory IgA; mucous membranes	Found in tears, saliva, GI and respiratory secretions. Those secretions protect mucous membranes, which are portals of entry. Genetic

		inability to make IgA → eye, respiratory, and GI infections.
IgM	First to appear; suggests current infection	Does not hang around. Hepatitis example: IgM against the virus = recent or current. IgG against the virus = they had it sometime (2 months, 2 years, or 10 years — you cannot tell).
IgD	On the B cell; stimulates proliferation	Not usually free in blood. Sits on the B-cell membrane like a monitor. If its antigen is still around and binds, the B cell proliferates and makes more antibody.
IgE	Allergic reactions; defends against parasites	In this country we mostly remember the allergy half because parasite disease is uncommon. She postponed the allergy detail to next class.

TEST TIP: IgM = now. IgG = sometime, and it lasts. IgA = secretions / mucosa. IgG is the placental one. IgE = allergy and parasites.

7. Active vs passive immunity

She said (not on the slide): Students mix these on the test. Her rule: active means YOUR immune system had to generate it. Passive means your immune system did nothing and just received it. **Her handout: both active and passive are types of adaptive immunity, not innate. Active is long-lasting but takes a few days to weeks after the first exposure. Passive is short-term only, and can be antibodies or immune cells received from another source.**

Example she walked	Which	Why
You get strep throat and make antibodies	Active	Your system made them
You get a vaccine	Active	The shot is antigen. You still have to make the antibody.

Mom's antibodies cross the placenta to the fetus	Passive	Mom made them, not the baby
Gamma globulin injection after a hep B needle stick	Passive	Somebody else made the antibody; employee health injects it

Slide line to memorize: gamma globulin (IgG) = antibody → passive immunity. Vaccine = antigen → active immunity.

She said (not on the slide): People mix them because both can be a shot. The difference is what is in the syringe.

	Passive	Active
Starts	Right away — protected as soon as the antibody arrives	Slow. Antibodies have to build up. COVID vaccine example: not protected until about 2 weeks later.
Lasts	Not long. Antibodies break down. Your system does not know how to replace them because it did not make them.	Long. When those antibodies age out, your system knows how to make more.

She said (not on the slide): The summary chart on the next slide is reused in third quarter adult health 1. She told you not to cram and dump it.

8. Immunodeficiency disorders

One or more branches of the immune system are not working.

Expected problems:

- Lots of infections: persistent, hard to treat, or caused by organisms that usually do not bother people.
- Higher cancer incidence, because cancer surveillance fails.

She said (not on the slide): *Severity depends on how many branches are hit. It can be one antibody class, all antibodies, B and T both, or only phagocytes, or only complement.* Her handout's five deficiency types: B-cell immunodeficiency; T-cell immunodeficiency; combined T-cell and B-cell immunodeficiency; disorders of the complement system (one or more components missing); disorders of phagocytosis (neutrophils, monocytes/macrophages).

Phagocyte failure patterns she listed:

- Can engulf but missing the enzymes to kill what was ingested.
- Can kill but lost chemotaxis, so they do not know where to go.
- Know where to go but lost mobility, so they cannot get there.

Complement failure: missing an activating factor → cannot generate inflammation, so one defense tool is gone. Missing an inhibiting factor → inflammation cannot be slowed once it starts.

Primary vs secondary

	Primary	Secondary
When	Born with it	Acquired later
Why	Inherited gene defect / mutation → defective white cells that cannot do the job	Infection (AIDS), cancer (leukemia), immunosuppressive drugs, or another organ failure that hits immunity Her handout's cancer example is lymphoma, and she also lists other diseases that affect synthesis or maintenance of immune cells or components.

She said (not on the slide): ***Liver makes complement factors, so liver failure***

compromises complement. Some kidney diseases damage the glomerular membrane so antibodies, normally too big to leak, spill into urine and are not in the blood where they protect you.

9. AIDS — the immunodeficiency she chose

DO NOT MIX UP: AIDS is an immunodeficiency disorder, not an autoimmune disorder. It is in the name: Acquired Immunodeficiency Syndrome. She said people say the acronym and then forget that.

Cause: HIV, human immunodeficiency virus. A retrovirus that selectively attacks the immune system, mainly helper T cells.

Lose helper T and you lose all adaptive immunity (no master switch) and innate immunity works less efficiently.

Opportunistic infection = infection by an organism that does not bother people with a normal immune system.

How HIV replicates

She said (not on the slide): Retrovirus has RNA only, not DNA. Most viruses have one or the other, not both.

Enters a host cell, usually a helper T.

Changes viral RNA into viral DNA. The enzyme on her handout is reverse transcriptase.

Inserts that viral DNA into the host cell's DNA.

Host DNA is the protein recipe, so the cell now makes more virus.

Virus numbers rise until the helper T breaks apart and new HIV is released to infect other helper T cells.

Early on the body replaces many killed helper T cells and kills many viruses. Over time helper T counts fall, virus rises, and immunity fails.

Three routes of transmission

Route	What she actually said
Blood / blood products	Not mainly transfusion. U.S. blood has been screened since the mid-1980s. A middle-of-the-road figure she used: about 1 in 2 million transfusions. She also cited about 4.5 million Americans a year who would die without a needed transfusion — so if you need blood, have it. Real blood route: shared needles among IV drug users. Healthcare needle sticks are rare but have transmitted it. Be careful with needles.
Sexual contact	Semen and vaginal secretions. Not casual contact: not a handshake, toilet seat, or shared glass.
Perinatally	Usually NOT in utero — placenta is a good barrier. Usually exposure to mom's blood in labor and delivery, or breastfeeding. Depends on mom's viral load.

Two stages

Stage	Definition she used
HIV infection	Infected, no immunodeficiency symptoms yet. Usually no symptoms until helper T counts are very low. Latent: virus is there, immune system keeps viral levels low. Can last up to 10 years or more even without antivirals; drugs may stretch it longer. Dangerous because they can still transmit. Her handout: in the latent phase the cell serves as a reservoir for the virus.
AIDS	Diagnosis switches when helper T count drops below 200, OR they develop an AIDS-defining illness. Normal helper T she gave: about 600 to 1,200.

TEST TIP: AIDS threshold: CD4 below 200, or an AIDS-defining illness. Normal frame: 600–1,200.

Window period

Time between infection and seroconversion (blood test flips from negative to positive). During it, nobody knows, so nobody is preventing transmission. That is the dangerous part of the latent phase.

She said (not on the slide): Length depends on the test. Textbook: 1 to 3 months with the cheaper screening tests used on large numbers of people. Blood banks use a more expensive test that shortens the window to about 2 or 3 weeks. Many hospital patients may not know they have HIV, so use standard precautions on everybody. She said this is not because patients lie; they may not know.

Tests

Test	Use
EIA (enzyme immunoassay; also called ELISA / enzyme-linked immunosorbent assay)	Common screen. Not HIV-specific — a technique also used for other viruses (she mentioned CMV). Has false positives, so a positive must be confirmed. She prefers false positives over false negatives on a screen so infected people do not fall through.
Western blot	Follow-up if EIA is positive. More expensive, more specific. Positive = HIV infection. Negative = the EIA was a false positive.
Home finger-stick mail-in	Lab runs both tests. Results by phone in about 3 to 7 days.
Home oral swab	You read it in about 20 minutes. EIA only. A positive still needs a Western blot.

Once it is AIDS

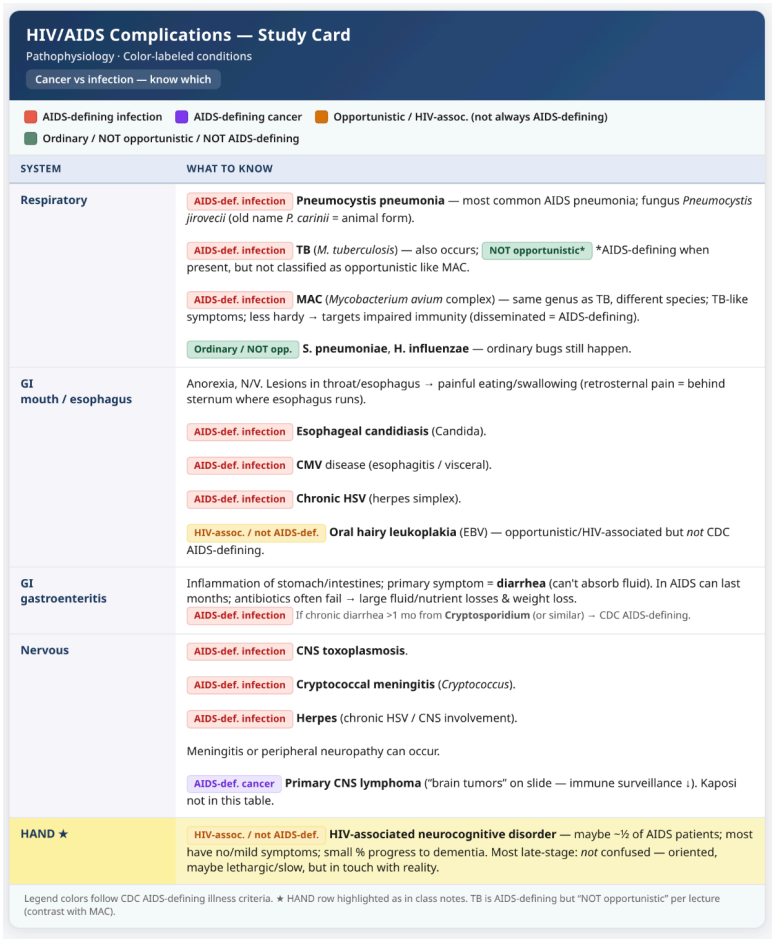
CDC AIDS surveillance case definition is a list of AIDS-defining illnesses, mostly opportunistic infections and cancers. Infections become more numerous, more severe, and harder to treat — both ordinary ones and opportunistic ones.

She said (not on the slide): Without antivirals, death in about 2 to 3 years. Antivirals extend that.

Assessment findings she reviewed

System	What to know (know whether cancer or infection)
Respiratory	Pneumocystis pneumonia is the most common AIDS pneumonia. Fungus name: Pneumocystis jirovecii (on the slide). Old name Pneumocystis carinii is the animal form. TB (Mycobacterium tuberculosis) also occurs and is NOT opportunistic. Related opportunistic cousin: Mycobacterium avium complex — same genus, different species, TB-like symptoms, not as hardy, so it hits impaired immunity. Ordinary bugs still happen: Streptococcus pneumoniae, Haemophilus influenzae.
GI — mouth and esophagus	Anorexia, nausea, vomiting. Lesions in throat and esophagus from Candida, CMV, herpes simplex. -> Painful eating and swallowing. Retrosternal pain = behind the sternum, where the esophagus runs. Epstein-Barr virus can cause oral hairy leukoplakia.
GI — gastroenteritis	Inflammation of stomach and intestines. Primary symptom is diarrhea because inflamed gut cannot absorb fluid. In AIDS it can last months, antibiotics often fail, large fluid and nutrient losses, weight loss.
Nervous system	Toxoplasmosis, Cryptococcus, herpes. Meningitis (meninges) or peripheral neuropathy. Brain

	tumors also occur because surveillance is down.
HAND (part of nervous)	HIV-associated neurocognitive disorder. Maybe half of AIDS patients. Diagram: most have no symptoms or only mild ones. Only a small percent progress to dementia. Most late-stage patients are not confused. Oriented, maybe a little lethargic or slow, still in touch with reality.



HIV/AIDS Complications — Study Card

Pathophysiology · Color-labeled conditions

Cancer vs infection — know which

AIDS-defining infection · AIDS-defining cancer · Opportunistic / HIV-assoc. (not always AIDS-defining) · Ordinary / NOT opportunistic / NOT AIDS-defining

SYSTEM	WHAT TO KNOW
Respiratory	AIDS-def. infection **Pneumocystis pneumonia** — most common AIDS pneumonia; fungus *Pneumocystis jirovecii* (old name *P. carinii* = animal form). AIDS-def. infection **TB** (*M. tuberculosis*) — also occurs; NOT opportunistic* *AIDS-defining when present, but not classified as opportunistic like MAC. AIDS-def. infection **MAC** (*Mycobacterium avium* complex) — same genus as TB, different species; TB-like symptoms; less hardy → targets impaired immunity (disseminated = AIDS-defining). Ordinary / NOT opp. **S. pneumoniae**, **H. influenzae** — ordinary bugs still happen.
GI mouth / esophagus	Anorexia, N/V. Lesions in throat/esophagus → painful eating/swallowing (retrosternal pain = behind sternum where esophagus runs). AIDS-def. infection **Esophageal candidiasis** (Candida). AIDS-def. infection **CMV** disease (esophagitis / visceral). AIDS-def. infection **Chronic HSV** (herpes simplex). HIV-assoc. / not AIDS-def. **Oral hairy leukoplakia** (EBV) — opportunistic/HIV-associated but *not* CDC AIDS-defining.
GI gastroenteritis	Inflammation of stomach/intestines; primary symptom = **diarrhea** (can't absorb fluid). In AIDS can last months; antibiotics often fail → large fluid/nutrient losses & weight loss. AIDS-def. infection If chronic diarrhea >1 mo from **Cryptosporidium** (or similar) → CDC AIDS-defining.
Nervous	AIDS-def. infection **CNS toxoplasmosis**. AIDS-def. infection **Cryptococcal meningitis** (*Cryptococcus*). AIDS-def. infection **Herpes** (chronic HSV / CNS involvement). Meningitis or peripheral neuropathy can occur. AIDS-def. cancer **Primary CNS lymphoma** ("brain tumors" on slide — immune surveillance ↓). Kaposi not in this table.
HAND ★	HIV-assoc. / not AIDS-def. **HIV-associated neurocognitive disorder** — maybe ~½ of AIDS patients; most have no/mild symptoms; small % progress to dementia. Most late-stage: *not* confused — oriented, maybe lethargic/slow, but in touch with reality.

Legend colors follow CDC AIDS-defining illness criteria. ★ HAND row highlighted as in class notes. TB is AIDS-defining but "NOT opportunistic" per lecture (contrast with MAC).

Cancers

Kaposi sarcoma: a vascular malignancy — cancer of cells lining some smaller blood vessels. Raised reddish-purple skin lesions, often trunk or head and

neck, can be on internal organs and then cause organ dysfunction. Looks different person to person. Head and neck lesions disfigure and hurt.
Cervical cancer and non-Hodgkin lymphoma also occur. Neither is opportunistic; both can happen with a normal immune system too.
Lymphoma

Wasting syndrome

She said (not on the slide): ***Skinny / wasting away is not enough. To count as wasting syndrome the weight loss has to happen WITHOUT an opportunistic infection or cancer that could explain it. In parts of Africa with a lot of AIDS it is called slim disease because wasting is so common.***

Interventions — two levels

Treat the infection or cancer that already happened with its standard therapy: antibiotics, antifungals, chemo.
Treat the underlying HIV with antivirals that slow replication — block entry into new cells, or interfere with enzymes the virus needs to copy itself.

She said (not on the slide): *Drugs have slowed progression and improved survival enough that AIDS is now treated as a chronic illness. New drugs now raise questions that did not matter when survival was short, such as whether a drug raises cholesterol and heart-disease risk. Undetectable viral level is not the same as cured. We cannot yet remove the viral DNA sitting in host-cell DNA, and that DNA can still make new virus. She will not say we have cured AIDS.*

10. Rapid-fire test bank

Four functions: immunity, self vs non-self, inflammation, cancer surveillance. Overlap is expected.
Antigen = marker, not automatically bad. Antibody = what we make against a foreign antigen.
Primary lymph organs: bone marrow and thymus. Secondary: spleen, nodes, tonsils, appendix, mucosal patches. Primary failure hurts more.
Innate = born with it, nonspecific, fast. Adaptive = acquired, specific, slower, stronger.
Five innate pieces: barriers, phagocytes/APCs, NK cells, cytokines, complement.
NK is the only innate lymphocyte. Kills with toxins, not phagocytosis.
Cytokine = white cell telling other white cells what to do. Fever is useful. We give synthetic cytokines and cytokine blockers.
Complement word = inflammation. Also chemotaxis, lysis, opsonization. Mast cell ≈ basophil; histamine, leukotrienes, bradykinin.
B cells / antibodies = outside the cell, usually bacteria. T cells = inside the cell, usually viruses, by killing the infected cell.
CD4 helper = master switch. CD8 cytotoxic needs helper T and is antigen-specific; NK does not and is not. Regulatory T stops the response and stops self-reactive cells. Memory T remembers; maybe more mitochondria.
Plasma cell = antibody. Memory B = fast second response.
IgG placenta and most plentiful. IgA secretions. IgM now. IgD on B cell. IgE allergy and parasites.
Vaccine = antigen = active. Gamma globulin = antibody = passive. Passive is fast and short. Active is slow and long. COVID example: about 2 weeks.
Immunodeficiency: infections plus cancer. Primary = born with it. Secondary = AIDS, leukemia,

immunosuppressants, liver (complement), kidney (antibody loss in urine).
AIDS is immunodeficiency, not autoimmune. HIV hits CD4. Retrovirus: RNA → DNA → into host DNA.
Routes: shared needles more than transfusion (~1 in 2 million). Sex, not casual contact. Perinatal usually at delivery or breastfeeding, not in utero.
Window: 1–3 months on cheap screens; 2–3 weeks at the blood bank. Standard precautions on everyone. EIA screen, Western blot confirms. Home oral swab is EIA only.
AIDS: CD4 < 200 (normal 600–1,200) or AIDS-defining illness. Without antivirals, about 2–3 years. PCP = P. jirovecii, most common AIDS pneumonia. MAC is opportunistic; TB itself is not. HAND rarely reaches dementia. Wasting syndrome has no explaining infection or cancer. Undetectable is not cured.

11. End of deck

Slides 85–93 are cartoons with no recorded audio. She closed with: enjoy the cartoons, see you in class.

Class 5 — Immunity & Inflammation, Part 2: Autoimmunity & Hypersensitivity

NRSG 224 Pathophysiology · Fall 2026 · filled from the class study guide

> **Personal study sheet — not for submission.** Questions keep the guide's order. Brackets show where each answer came from: S = recorded-lecture slide, mm:ss = recording time (1x), sheet p. = your full study sheet. not covered in lecture = textbook knowledge, not from her lecture.

Sources: recorded lecture subtitles (times are the 1x recording, about ±30 s; on the 2x video, halve them), recorded and class slides, verified clinical excerpts, and your full study sheet pp. 17–19.

Page 1 — Distinguishing self from nonself

Guide: "Self-tolerance (the absence of an immune response directed against a person's own antigens) normally is maintained through:" — two blank dashes, then MHC molecules (3rd mechanism, already printed).

A1 Self-tolerance mechanism #1 (blank –)

- Mechanism #1 = regulatory T cells (aka suppressor T cells) [02:13]
- Two jobs: shut off immune response once pathogen is gone; suppress any potentially self-reactive WBCs [02:13]
- Gets rid of those self-reactive cells before they can engage in autoimmune activity [02:48]
- Sheet version: turn off immune response + suppress self-reactive WBCs [sheet p.17]

A2 Self-tolerance mechanism #2 (blank –)

- Mechanism #2 = bone marrow + thymus screening of immature WBCs [02:48]
- T cells mature in thymus, B cells mature in bone marrow; both sites screen for potentially self-reactive cells [02:48]
- If found → get rid of self-reactive cells BEFORE release into general circulation [03:20]
- Sheet version: screen + remove immature self-reactive B cells and T cells [sheet p.17]

Page 1 — Inflammation

B1 S/S (cardinal signs): list all 5 (fills the five blank dashes) and for each, the mechanism she gave

- Redness + heat ← vasodilation → more blood to area [58:27]
- Swelling (edema) ← ↑ capillary permeability → fluid leaks into tissues [59:31]
- Pain ← swelling; usually goes along with it [59:31]
- Impaired/loss of function ← swollen parts don't work (e.g., sprained finger can't bend) [1:00:02]
- Cardinal signs listed: redness, heat, swelling, pain, impaired function [sheet p.18]

B2 Vascular response: momentary vasoconstriction, then vasodilation → ___

- Momentary vasoconstriction → vasodilation [sheet p.18]
- Vasodilation = first vascular event → brings more blood: O2, fuel, antibodies to fight

infection [58:27]

- Result = the redness and warmth of inflammation [58:59]

B3 ↑ capillary permeability → fluid escapes into tissues → ___

- ↑ capillary permeability = capillaries get 'leaky' → fluid leaks into tissues, bringing nutrients + antibodies [59:31]
- Causes edema, pain, and loss of function [59:31]
- Exudate types: serous (clear, light yellow = serum), serosanguineous (watery blood/pink), purulent (pus = dead WBCs, infected) [1:01:08]; [slide 53]
- Exudate is usually GOOD: brings nutrients/O2/antibodies, flushes dirt, dilutes irritants, slows spread of contaminants [1:02:13]

B4 Cellular response – movement of WBCs into area of injury → ___

- Cellular response = phagocytic WBCs migrate to the injury area [1:03:53]
- Neutrophils get there first, followed by monocyte-macrophages [1:03:53]
- They phagocytize pathogens + dead tissue → get it ready for repair [1:03:53]
- If needed, they act as APCs → signal/activate adaptive immunity [1:04:24]

B5 Acute-phase response (fever, ↑ # WBCs, skeletal muscle catabolism) — what to know

- Acute-phase response = systemic effects: fever, leukocytosis, ↑ inflammatory markers, varying degrees of skeletal muscle catabolism [slide 57]; [1:05:39]
- Fever is beneficial: bacteria can't survive higher temp + WBCs become more active [1:05:26]
- Leukocytosis = actual ↑ in number of WBCs; fever + ↑WBC ≠ always infection (post-op incision, MI) → monitor trends, check pus, wound culture [1:05:39]
- Skeletal muscle catabolism = break down muscle protein → amino acids to make antibodies / repair tissue; body mobilizes only what's useful (mosquito bite vs appendicitis) [1:10:21]
- Markers: CRP (liver; regulates cytokines), fibrinogen (helps locate pathogens), ESR (markers stick to RBCs → heavier → sediment faster) [36:36]; [37:37]
- ★ **Prof emphasis:** 'those are the only three I'm going to expect you to recognize for the test' (CRP, fibrinogen, ESR) [39:49]

B6 Potential problems r/t inflammation

- Inflammation can get out of control: very severe, or recurrent/persistent → harmful [1:12:01]
- Chronic irritation/inflammation → metaplasia + scar tissue formation [1:12:34]

- Scar tissue → loss of function + deformity (RA joints pulled into abnormal positions) [1:13:09]
- Kidney scarring → can't filter blood / make urine like real kidney tissue [1:13:41]

B7 Potential problems if immune system function impaired/altered

- Immunodeficiency → infection [class slide 4]
- Hypersensitivity → allergic reactions → unnecessary inflammation [class slide 4]
- Autoimmunity → damage to self tissue [class slide 4]
- Diagnostics: immunodeficiency → cultures, WBC count; hypersensitivity → allergy testing; autoimmunity → check for autoantibodies [class slide 5]
- ★ **Prof emphasis:** ALL three can cause inflammatory markers to increase [class slide 5]

Flowchart 1 — Autoimmunity (breakdown in self-tolerance)

C1 Nursing problem:

- Nursing problem: damage to self tissue — immune system attacks self-antigens [class slide 4]
- Injured self cells → organ/body system not working properly [22:17]
- Most autoimmune diseases are chronic: cycles of remissions and exacerbations/flares [32:40]
- Goal of care: minimize exacerbations, get patients into remission, keep them there as long as possible [33:14]

C2 SLE — Assessment findings:

- Ag-Ab complexes deposit in collagen-rich tissues → many body systems; skin: bruising, rashes, classic butterfly/malar rash (worsened by sun; flares triggered by sun) [46:46]; [47:19]
- Joints (lupus arthritis), kidney (inflammation, renal failure), lungs, heart, eyes, GI tract; not all systems in every patient [48:01]
- Autoantibodies: ANA (anti-nuclear) is a primary one — antibodies vs nuclear/DNA antigens; also autoantibodies vs RBCs and platelets [43:32]; [45:34]
- Labs: ↑ inflammatory markers (CRP, fibrinogen, ESR) = inflammation only; autoantibody list on slide also Anti-DNA, Anti-CCP, Anti-phospholipids, RF [slide 34]; [35:28]
- Dx is a diagnosis of exclusion — 'the great imitator'; no one definitive test; can take years [48:43]

C3 Interventions (autoimmune / SLE):

- Two levels of intervention: stop the underlying autoimmune attack + treat the symptoms [41:39]; [slide 35]
- Stop attack: immunomodulators — immunosuppressive drugs, monoclonal antibodies, corticosteroids; also plasmapheresis [41:58]; [slide 44]
- Treat symptoms/inflammation: NSAIDs (Motrin, ibuprofen, aspirin) — fewer side effects, less powerful; corticosteroids = synthetic cortisol, powerful but many side effects [51:28]

- Use NSAIDs first, add corticosteroids if necessary (NSAIDs won't halt kidney damage) [52:31]
- Keep lupus patients out of the sun — sun exposure triggers exacerbations [47:19]

Flowchart 2 — Hypersensitivity disorders

D1 Nursing problem:

- Problem: immune system reacts to an antigen most people would not react to; not normal/appropriate, usually excessive → can damage host tissue [1:14:17]
- Results in unnecessary inflammation / allergic reactions [class slide 4]
- Hypersensitivity = broad umbrella; allergy (reacts to environmental antigen, e.g., cat dander) and autoimmune disease (reacts to self-antigen) are two specific types [1:14:50]
- Predisposing factors: BOTH genetic predisposition + exposure to triggering antigen (inhaled, ingested, injected, skin contact) [1:16:20]

D2 Type I — allergic rhinitis (hay fever)

- Type I = IgE-mediated; allergic rhinitis = 'hay fever' [1:24:49]
- Findings: runny nose, lots of sneezing, red/itchy/watery eyes [1:24:49]
- Mechanism: mast cells/basophils degranulate in nasal passages / eye mucous membranes [1:27:27]

D3 Type I — asthma

- Type I, IgE-mediated; airway constriction [1:25:12]
- Findings: wheezing, shortness of breath [1:25:12]
- Mechanism: degranulation in the airways [1:27:27]
- Reaction within ~15 min of trigger exposure (e.g., cat-allergic person starts to wheeze in ~15 min) [1:37:56]

D4 Type I — urticaria

- Urticaria = hives; Type I, IgE-mediated [1:25:43]
- Looks like mosquito bites all over the body; very itchy [1:25:43]
- Mechanism: degranulation in the skin [1:27:27]

D5 Type I — atopic dermatitis (eczema)

- Atopic dermatitis = eczema; Type I, IgE-mediated [1:25:12]
- Findings: itchy, scaly, crusty rashes — 'incredibly itchy' [1:25:12]
- Mechanism: degranulation in the skin [1:27:27]

D6 Type I — anaphylaxis

- Most severe Type I disorder; main worry = blood pressure drops → hypotension → can progress to anaphylactic shock; people can die [1:26:14]
- Also bronchoconstriction and some shortness of breath [1:26:14]
- Mechanism: degranulation system-wide → whole-body vasodilation → BP drop [1:27:48]

- Related: angioedema = tongue, lips, part of face swell [1:25:43]
- Textbook, not from her lecture: anaphylaxis emergency care = IM epinephrine first, then airway/oxygen, IV fluids for hypotension, plus antihistamines/corticosteroids

  [not covered in lecture — textbook] not covered in lecture

D7 Type II — blood transfusion reactions

- Type II = antibody-mediated/cytotoxic; uses IgG and IgM (not IgE) [1:30:28]
- Example: type A recipient (has antibodies vs B antigen) given type B blood → recipient antibodies attack the donor cells [1:31:00]
- When antibody binds antigen on donor cell → destroys the donated cell; the binding itself causes the damage [1:31:00]

D8 Type III — SLE (why it is Type III)

- Type III = immune complex-mediated (IgG/IgM); immune complex = antigen-antibody complex [1:32:10]
- Not the binding itself — complexes get deposited in tissue → activate complement → inflammation = the symptoms [1:33:09]
- That is exactly the lupus scenario (ANA binds DNA antigen → complexes deposit in collagen-rich tissue) [44:33]; [1:34:13]

D9 Type IV — contact dermatitis (poison ivy), diabetes mellitus

- Type IV = T-cell-mediated / cellular — only type NOT antibody-mediated; delayed [1:34:44]

- T cells get sensitized to antigen → next exposure, T cells attack → symptoms [1:34:44]
- Examples: poison ivy/poison oak contact dermatitis (rash where touched); type 1 diabetes (sensitized T cells attack pancreatic cells); certain transplant rejections [1:35:16]
- Delayed timing: rash/peak ~1–3 days after exposure (vs Type I/II 15–30 min, Type III ~6 hr) [1:37:56]

D10 Assessment findings (hypersensitivity) — guide says 'Specific to individual disorders'

- Findings are specific to individual disorders; symptoms differ by WHERE mast cells/basophils degranulate (airways→asthma, nose/eyes→rhinitis, skin→eczema/hives, system-wide→anaphylaxis) [1:27:15]
- Diagnostics: allergy testing — skin testing, sticking little bits of antigen under the skin to see which cause a reaction [1:29:22]; [class slide 5]
- All three major immune problems can increase inflammatory markers [class slide 5]
- ★ **Prof emphasis:** on the test she can't ask Type II vs Type IV from symptoms; but anything that sounds like an 'allergy' = Type I IgE-mediated [1:36:33]

D11 Interventions (hypersensitivity):

- Main interventions: get rid of/avoid the trigger + treat the symptoms; we can't desensitize once a person is sensitized [1:39:01]
- Meds for Type I: antihistamines (histamine is a mediator) + anti-inflammatories —

corticosteroids local vs systemic (e.g., fluticasone), others [1:29:05]; [slide 70]

- Help patients identify allergens (skin testing) and avoid them; if unavoidable (common trees, pets) → rely more on medications [1:29:53]
- ★ **Prof emphasis:** we can NEVER stop monitoring patients for allergic reactions to meds — B cells may start making IgE after the 1st, 10th, or 50th exposure (penicillin story) [1:23:32]

NRSG 224 — Circulatory Part 1 (Exam Notes)

High-yield only. Red = spoken detail she said that was not on the slides. Black = slides + handout.

Circulatory system overview

Three components: heart (pump), blood vessels (carry), blood (fill the vessels). Blood was already covered; this unit is heart and vessels.

Primary function: transport. What it moves:

- O2 and nutrients to cells (for energy)
- Waste away from cells to organs of elimination (lungs, kidneys)
- Electrolytes and hormones (cell function)
- Immune substances / antibodies (defense)
- Heat from core to skin (thermal regulation)

If even one of those fails (example: no O2/nutrient delivery), cells die. That is why a bad circulatory system causes so many problems.

Heart chambers and what fails when they fail

Four chambers: two atria on top, two ventricles on bottom.

Atria — two jobs:

- Collect blood returning to the heart
- Serve as pumps that fill the ventricles

If atria do not fill the ventricles → not much blood goes out per beat → cardiac output drops.

Second problem if atria fail: blood pools and stagnates in the atria → blood wants to clot. Those clots can drain into the ventricles, get pumped out, occlude arteries elsewhere, and cause ischemia and tissue damage.

Ventricles = main pumping chambers. If they fail, cardiac output drops.

Right ventricle pumps to the lungs = pulmonary circulation. Right ventricle fails → blood is not oxygenated → poor oxygenation.

Left ventricle pumps to the rest of the body = systemic circulation. Left ventricle fails → oxygenated blood does not reach the tissues → major organs can fail.

Valves: keep blood going the right direction, not backwards. Pericardium: sac that holds the heart in position in the chest and protects it. Myocardium: thick muscle layer that does the work. Endocardium: thin inner lining. (More detail next class.)

Cardiac cycle and cardiac output

Work of the heart = contract and relax, pumping blood. That cycle = cardiac cycle.

- Systole = contracts
- Diastole = relaxes and fills

Cardiac output = amount of blood pumped out of the heart per minute.

CO changes through the day with body needs. Sitting through a lecture needs little CO; playing soccer needs a lot.

CO = HR × SV

Why the equation works: number of beats per minute × amount of blood per beat = amount of blood per minute.

Faster HR → higher CO, only up to a point. If HR is too fast, the ventricles do not have time to fill between beats and CO drops. That point is roughly 150–180–200, depending on the person's fitness.

HR is mostly set by the autonomic nervous system: the balance of sympathetic vs parasympathetic activity.

Stroke volume: preload, afterload, contractility

SV = blood pumped out per beat. Determined by preload, afterload, and myocardial contractility.

Preload

- Amount of blood in the ventricle ready to be pumped out on the next beat.
- Determined by venous return. More return → more in the ventricle → more stretch of the muscle fibers → stronger contraction → bigger SV.
- Frank-Starling law (Starling's law of the heart): more blood back → more stretch → stronger force → bigger SV. Elastic analogy: more stretch, harder snap-back.
- Limit: maximum CO when fibers are stretched about 2½ times resting length. Stretch more than that and force drops — like an overstretched elastic waistband that no longer snaps back. Too much blood in the ventricles → less SV → less CO.

Afterload

- Resistance the heart must pump against to eject blood. Determined mostly by arterial blood pressure.
- Higher BP → more resistance → less blood out per beat → lower SV → lower CO.
- Bicycle-tire analogy: flat tire is easy to pump and a lot of air goes in per stroke; as pressure rises, each stroke moves less. Same for the heart ejecting into high-pressure arteries.

Cardiac contractility

- Force of ventricular contraction that is independent of preload and afterload. Due to actin–myosin interaction.
- Inotropic influences: positive increases force, negative decreases force.

- Positive inotrope example: sympathetic (fight-or-flight) stimulation. Heart rate rises and the heart pounds because force and SV also rise.
- Negative inotrope examples: hypoxia (heart muscle cannot make enough ATP for a strong contraction → weak contraction → low SV → low CO); heart attack (dead area of wall does not contract with the rest → force drops → SV and CO drop).

Why CO matters: without CO, none of the transport functions happen. Most cardiac diseases cause problems because CO drops. Interventions aim to raise CO — increase preload, decrease afterload, or increase contractility.

Blood vessels and the force that moves blood

Three kinds: arteries (heart → body), capillaries (gas and nutrient exchange), veins (body → heart).

For blood to flow you need two things: patent vessels (clear, not clogged) and a force to move the blood. That force is arterial blood pressure. If BP is too low, there is no force and circulation is poor.

Normal BP is about 120/80.

BP (mean arterial) = CO × PVR

More CO → higher BP. Phone-booth / Volkswagen-bug analogy: the more people you stuff into a fixed space, the higher the pressure. Same for blood in arteries.

PVR is set mostly by vessel diameter. Dilated vessels = more space = lower pressure. Constricted vessels = same blood in a smaller space = higher pressure.

Green items on her slide are the ones easiest to change with drugs or other interventions.

- BP too low: start an IV and give fluid (raise volume) or give a vasoconstrictor (smaller diameter → higher PVR → higher BP).
- BP too high: diuretic (lower volume), drug to slow HR (lower CO → lower BP), or vasodilator (lower PVR → lower BP).

This is why the BP-drug categories in pharmacology make sense once you know the equation.

What controls HR, volume, and vessel diameter

Three primary controllers: autonomic nervous system, kidneys, ADH.

Autonomic nervous system

- Balances sympathetic and parasympathetic to keep HR and BP in range.
- Affects HR and vessel diameter. Does not affect blood volume.
- Two centers in the medulla: vasomotor center (sympathetic effects) and cardioinhibitory center (parasympathetic effects). Together they set HR and vessel diameter.

Kidneys

- Make urine → control blood volume.
- Also run the renin-angiotensin-aldosterone system. If BP/volume drops → blood flow to the kidney drops → kidney releases renin → renin converts angiotensinogen to angiotensin I → angiotensin I is converted (in the lung) to angiotensin II → angiotensin II constricts vessels and triggers aldosterone → aldosterone holds sodium and water → volume and BP rise.

ADH (antidiuretic hormone)

- Released from the posterior pituitary when volume or BP is low.
- Vasoconstricts and holds water → volume and BP rise.

Hypertension

Primary (essential): no identifiable disease causing it. Secondary: caused by another disease (example: renal disease).

Primary is not a specific disease of one system. The three control systems (ANS, kidneys, ADH) have shifted their set point — they have been reprogrammed to hold BP at a higher level. That is not the same as a disease process; a disease process would be secondary HTN.

Guidelines were updated in 2017. Normal requires both systolic <120 and diastolic <80. If only one number is high, you are already out of normal.

Old diagnosis threshold was chronically >140/90. Under the current guidelines, 140/90 is already stage 2. The old pre-hypertension category was dropped because people heard pre and thought they did not have it yet. Stage 1 and stage 2 use or, not and: only one number has to fit the category.

Diagnose only if elevated on two or more occasions. One high reading is not enough.

Silent killer: no symptoms other than the high number itself. Occasionally headaches or nosebleeds if the pressure is extremely high, but most people feel nothing.

We care because HTN predisposes to atherosclerosis, which causes heart disease and stroke.

Two ways HTN promotes atherosclerosis:

- Mechanical stress → arterial walls hypertrophy (thicken) to withstand the pressure → lumen narrows and the wall becomes less elastic (hardened).
- Mechanical stress damages the endothelial lining.

Predisposing factors to HTN

Non-modifiable

- Family history
- Age (older; but HTN is appearing younger because modifiable risks are rising in the population)
- Race: highest in Black / African American patients — more frequent, more severe, and at a younger age. Reason is unclear; possible salt sensitivity or higher sympathetic activity.

- Sex: men higher risk until menopause; after menopause women equal men. She keeps sex in non-modifiable.

Modifiable

- Diet: high sodium (holds water → volume up); low potassium
- Obesity, especially upper-body / apple shape
- Sedentary
- Excess alcohol: more than 2 drinks/day for men, more than 1 for women. You cannot bank the week's drinks for the weekend — that is binge drinking, which also raises cancer, accident, and violence risk.
- Stress: acute stress raises BP through sympathetic vasoconstriction, but primary HTN is chronic. Current evidence points to a social component (poverty, abuse, racism, sexism). Assess anxiety, depression, and stress.
- Smoking, high cholesterol, high fat are in parentheses because they associate more strongly with heart disease than with HTN, but they matter because HTN raises heart-disease risk.

Endothelium and atherosclerosis

Endothelium is the smooth inner lining of arteries. It used to be thought of as only a slippery surface. It actually produces substances that control constriction, dilation, and platelet aggregation.

Injury to endothelium → two consequences:

- Macrophages stick to the injured area, ingest LDL, become foam cells, squeeze between endothelial cells, migrate under the lining, and die there. Their material stays in the wall.
- Endothelium changes what it releases: less antithrombotic and vasodilatory substance, more growth factors → smooth muscle proliferates → fibrous cap forms over the plaque.

Once the plaque grows, three problems:

- It takes up space and narrows the lumen → less flow.
- The fibrous cap can rupture → clot forms on the rough surface → sudden occlusion.
- Pieces of plaque or clot can embolize downstream.

DO NOT MIX UP: Arteriosclerosis = general hardening/thickening of arteries. Atherosclerosis = the specific plaque form with lipid core and fibrous cap.

Predisposing factors to atherosclerosis

Non-modifiable: family history, age, sex.

Modifiable: smoking, HTN, diabetes mellitus, hypercholesterolemia.

Also associated (more with HTN, but assess them here too): upper-body obesity, sedentary, stress, poor sleep.

Cholesterol

- LDL carries cholesterol to tissues → available for plaque. Want low.
- HDL clears cholesterol from blood and returns it to the liver → less available for plaque. Want high.

- Guidelines: total <200 (experts now lean toward <170 or even <150 in high-risk patients); HDL >40 (below 40 = high CHD risk; ≥60 is considered cardioprotective and some experts let you subtract a risk factor); LDL <100 (some aim <70 in very high risk, but that usually needs medication and the side-effect trade-off has to be weighed); triglycerides <200 (some aim <150).

Ischemia vs infarction and collateral circulation

Ischemia = not enough arterial blood flow / oxygen, but no permanent damage yet. Infarction = ischemic necrosis; the tissue is dead.

Collateral circulation: alternate small vessels that can widen and supply an area if the main artery is blocked. Present in heart, brain, and large muscle groups (legs).

Collaterals develop more easily when the blockage is gradual (atherosclerosis). That is one reason atherosclerosis is often silent until plaques are large. A sudden clot leaves no time for collaterals to open, so an acute problem is more likely.

Interventions: lifestyle modification first (stop smoking, control diabetes, control HTN, lower cholesterol). Medications if lifestyle is not enough. Goal is to taper medications as lifestyle changes take hold.

Coronary heart disease / coronary artery disease

Atherosclerosis in the coronary arteries. She prefers CAD because it names the location: impaired flow through the coronary arteries.

Heart muscle does not get its oxygen from the blood inside the chambers. It gets oxygen from blood delivered through the coronary arteries. If those arteries fail, heart muscle is damaged and function drops.

Most common cause is atherosclerosis, so charts sometimes say atherosclerotic heart disease (ASHD). Predisposing factors are the same as for atherosclerosis.

Three ways patients present:

- Asymptomatic — plaques present, no symptoms yet. Least often seen, because they have no reason to come in.
- Chronic ischemic heart disease: stable angina, variant angina
- Acute coronary syndrome: unstable angina, acute myocardial infarction

Angina types

Stable (exertional / classic / typical)

- Triggered by increased demand (exertion, emotion, cold, large meal).
- Relieved by rest or nitroglycerin.
- Pattern is predictable.

Variant (vasospastic / Prinzmetal's)

- Coronary artery spasm, not a fixed plaque.
- Can occur at rest, often at night or early morning.
- Not reliably related to exertion.

Unstable (preinfarction / crescendo)

- Plaque is unstable. Pain at rest or with minimal exertion, longer, more frequent, or new.
- Not reliably relieved by rest or NTG. Warning that infarction may be next.

Levine's sign: clenched fist over the chest. Classic pain is substernal and may radiate to the left shoulder and down the inside of the left arm.

TEST TIP: Stable = predictable, demand-related, relieved by rest/NTG. Unstable = rest or minimal activity, not relieved, progressing. Variant = spasm, often at rest.

Acute myocardial infarction

Heart muscle cell injury or death → three consequences:

- Decreased cardiac output
- Inflammation → fever, elevated WBC
- Elevated cardiac markers (CK, LD, myoglobin, troponin)

Chest pain is usually more severe than angina and often needs opioids.

Nitroglycerin

White pill in a brown bottle, under the tongue. Vasodilator.

- Dilates arteries → lower BP → lower afterload → less workload
- Dilates veins → less venous return → lower preload → less workload

Bottom line: it works by decreasing demand, not by increasing supply.

3 × 5 rule: up to 3 doses, 5 minutes apart. If pain is still there after 3, treat as possible MI.

Revascularization and other treatments

PTCA (percutaneous transluminal coronary angioplasty): balloon opens the narrowed segment.

CABG: bypass the blocked segment, often with 3–4 grafts.

Low-dose aspirin: baby aspirin daily or adult aspirin every other day. Antiplatelet. Used to prevent a first or second MI. Avoid if gastric ulcers or aneurysms (bleeding risk).

Only for MI — thrombolytic / fibrinolytic therapy:

- Not the same as an anticoagulant (heparin, Coumadin). Anticoagulants keep existing clots from growing and prevent new clots; they cannot break a clot. A thrombolytic breaks the clot.

- Goal: reopen the artery and save cells that are injured but not yet dead. Dead cells cannot be saved.
- Start within 12 hours of symptom onset; maximum benefit within 2 hours.
- Not for patients at high risk of bleeding complications.

Only for MI — opioid analgesia (morphine):

- MI pain is severe enough to need opioids.
- Severe pain drives sympathetic stimulation → tachycardia → higher heart workload → worse supply-versus-demand mismatch. Relieving the pain removes that extra demand.

TEST TIP: If she asks why a drug is used, map it to the equation: does it change HR, preload, afterload, contractility, volume, or vessel diameter? That is how she framed the interventions.